

Viral-mediated Oct4 overexpression and inhibition of Notch signaling synergistically induce neurogenic competence in mammalian Müller glia

Reviewed Preprint

v2 • April 28, 2025

Revised by authors

Reviewed Preprint

v1 • April 2, 2025

Nguyet Le, Sherine Awad, Isabella Palazzo, Thanh Hoang, Seth Blackshaw 

Department of Neuroscience, Johns Hopkins University School of Medicine, Baltimore, United States • Department of Ophthalmology and Visual Sciences, University of Michigan School of Medicine, Ann Arbor, United States • Department of Cell and Developmental Biology, University of Michigan School of Medicine, Ann Arbor, United States • Michigan Neuroscience Institute, University of Michigan School of Medicine, Ann Arbor, United States • Department of Ophthalmology, Johns Hopkins University School of Medicine, Baltimore, United States • Department of Neurology, Johns Hopkins University School of Medicine, Baltimore, United States • Institute for Cell Engineering, Johns Hopkins University School of Medicine, Baltimore, United States • Kavli Neuroscience Discovery Institute, Johns Hopkins University School of Medicine, Baltimore, United States

 https://en.wikipedia.org/wiki/Open_access
 Copyright information

eLife Assessment

This manuscript demonstrates that Oct4 overexpression synergizes with Notch inhibition (Rbpj knockout) to promote the conversion of adult murine Müller glia (MG) into bipolar cells. These findings are **important** as the authors used rigorous genetic lineage tracing (GLAST-CreER; Sun-GFP) to confirm that neurogenesis indeed originates from MGs, addressing a key issue in the field. The single-cell multiomic analyses are **compelling**, and while functional studies of MG-derived bipolar cells would strengthen the conclusions, they are beyond the scope of this study.

<https://doi.org/10.7554/eLife.106450.2.sa3>

Abstract

Retinal Müller glia in cold-blooded vertebrates can reprogram into neurogenic progenitors to replace neurons lost to injury, but mammals lack this ability. While recent studies have shown that transgenic overexpression of neurogenic bHLH factors and glial-specific disruption of NFI family transcription factors and Notch signaling induce neurogenic competence in mammalian Müller glia, induction of neurogenesis in wild-type glia has thus far proven elusive. Here, we report that viral-mediated overexpression of the pluripotency factor *Oct4* (*Pou5f1*) induces transdifferentiation of mouse Müller glia into bipolar neurons, and synergistically stimulates glial-derived neurogenesis in parallel with Notch loss of function. Single cell multiomic analysis shows that *Oct4* overexpression leads to widespread changes in gene expression and chromatin accessibility, inducing activity of both the neurogenic transcription factor Rfx4 and the Yamanaka factors Sox2 and Klf4. This study demonstrates that viral-mediated overexpression of Oct4 induces neurogenic competence in

retinal Müller glia, identifying mechanisms that could be used in cell-based therapies for treating retinal dystrophies.

Introduction

The homeodomain factor Oct4 (Pou5f1) is a central component of transcriptional regulatory networks that maintain pluripotency in embryonic stem (ES) cells. Along with Sox2, Klf4, and Myc, Oct4 is one of the Yamanaka factors sufficient to induce the formation of induced pluripotent stem cells (iPSCs) from virtually all somatic cell types (Takahashi and Yamanaka, 2006). In contrast to these other factors, *Oct4* is selectively expressed in ES cells and essentially absent from somatic cells, including adult stem cells from every tissue examined. The two clear exceptions are found in gonadal stem cells (Takashima et al., 2013; Uhlen et al., 2010) and during neural crest formation, where *Oct4* expression is essential for the ability of these cells to generate diverse cell types (Morrison and Brickman, 2006; Zalc et al., 2021). This has raised the question of whether *Oct4* overexpression *in vivo*, alone or in combination with other pluripotency factors, may confer some level of multipotency on somatic cells in the central nervous system.

Sustained overexpression of the Yamanaka factors *Oct4*, *Sox2*, *Klf4*, and *Myc* induces teratoma formation in many tissues (Abad et al., 2013; Ohnishi et al., 2014). Transient or low-level Yamanaka factor overexpression, in contrast, has been reported to induce cellular rejuvenation in both neurons and other cell types, although this occurs without clear induction of proliferation or conferring multipotency (Ocampo et al., 2016; C. Wang et al., 2021). However, several studies have directly investigated whether viral-mediated overexpression in CNS neural stem or glial cells of Oct4 alone, or in combination with subsets of other pluripotency factors, can induce multipotency and neurogenesis. Results from these studies are mixed, with *Oct4* overexpression in neural stem cells reported to either not induce neurogenesis (Asadi et al., 2015; Dehghan et al., 2015; Sim et al., 2011), to promote formation of oligodendrocyte progenitors (OPCs) (Yu et al., 2021), or to promote remyelination in committed OPCs (Dehghan et al., 2016). Two studies reported that viral-mediated *Oct4* overexpression, either alone (Niu et al., 2013) or in combination with other Yamanaka factors (Gao et al., 2016), could directly convert reactive astrocytes into neurons. However, these studies used the GFAP minipromoter to drive gene expression, which has been reported to show insert-dependent ectopic insert-dependent silencing in glia and activation in neurons (L.-L. Wang et al., 2021). This limitation makes these findings difficult to interpret, as additional methods of validating cell lineage were not used.

Two previous studies have reported injury-dependent induction of *Oct4* in retinal Müller glial cells in zebrafish and mice. In zebrafish, where injury induces Müller glia to undergo conversion to neurogenic progenitors (Wan and Goldman, 2016), it was reported that *Oct4* is essential for glial reprogramming to occur (Sharma et al., 2019). In mice, where neurogenic competence is rapidly and actively suppressed following injury, it was reported that *Oct4* expression is transiently induced, but fully repressed within 24 hours following injury (Reyes-Aguirre and Lamas, 2016). This raises the possibility that *Oct4* expression may be sufficient to confer neurogenic competence on retinal Müller glial cells. In this study, we assessed whether AAV-mediated overexpression of Oct4 could induce reprogramming of adult mouse Müller glia. While we did not detect any expression of *Oct4* in either zebrafish or mouse Müller glia following injury, we used both genetic cell lineage and single-cell omics analyses to unambiguously demonstrate that viral-mediated overexpression of *Oct4* selectively induces the formation of bipolar neurons from control *GlastCreER;Sun1-GFP* mouse Müller glia without inducing substantial levels of proliferation. We further showed that *Oct4* overexpression synergistically enhances existing levels of Müller glia-derived neurogenesis in the absence of Notch signaling. Single-cell multiomic analysis demonstrated that Oct4 cooperates with Sox2 and Klf4, which are constitutively expressed in Müller glia, to broadly alter the gene regulatory landscape and induce expression of genes such as

the neurogenic transcription factor *Rfx4*. This study demonstrates that substantial levels of *in situ* glia-to-neuron reprogramming in mammalian retina can be induced by viral overexpression of a Yamanaka factor.

Results

***Oct4* and *Nanog* are not detectably expressed in Müller glia or Müller glia-derived progenitor cells in either zebrafish or mice**

To examine the expression of *Oct4* and other pluripotency-associated genes in Müller glia, we first analyzed previously published gene expression data from zebrafish and mouse retina (Hoang et al., 2020; Le et al., 2024). We first analyzed both single-cell RNA-Seq data obtained from wildtype retinas and bulk RNA-Seq data obtained from FACS-isolated Müller glia carrying glial-specific GFP reporter lines. Consistent with previous findings (Nelson et al., 2011; Surzenko et al., 2013), in mice we detected expression of *Sox2* and *Klf4* in resting and both NMDA-treated and light-damaged Müller glia, along with injury-induced expression of *Myc*, which is lost as activated glia progressively return to a resting state by 72 hours post-injury.

Likewise, in zebrafish we observed expression of *sox2* and *myca/b* (but not *klf4*) in both resting and injured Müller glia (Fig. S1a,b). However, we did not detect expression of either *Oct4* or its direct target *Nanog* in either species in any treatment condition (Loh et al., 2006). Bulk RNA-Seq data obtained from purified Müller glia detected very low levels of *Oct4* (*pou5f3*), but not *Nanog*, expression in some injured zebrafish Müller glia, but neither *Oct4* nor *Nanog* was detected in either resting or injured mouse Müller glia (Fig. S1a,b). In addition, we did not observe any induction of *Oct4* or *Nanog* in snRNA-Seq data obtained from mouse Müller glia following selective loss of function of *Rbpj* alone or in combination with *Nfia/b/x* (Le et al., 2024) (Fig. S1c). These results indicate that neither NMDA excitotoxicity, light damage, or induction of neurogenic competence by loss of function of *Nfia/b/x* and/or *Rbpj* induces detectable levels of *Oct4* or *Nanog* in neurogenic mammalian Müller glia.

AAV-mediated overexpression of *Oct4* induces conversion of control *GlastCreER;Sun1-GFP* mouse Müller glia to bipolar-like cells

To determine whether *Oct4* overexpression could reprogram mammalian Müller glia into a neurogenic state, we administered intraperitoneal tamoxifen injections into P21 *GlastCreER;Sun1-GFP* mice, thereby irreversibly labeling Müller glia and their progeny with nuclear envelope-targeted Sun1-GFP under control of the ubiquitous CAG promoter (de Melo et al., 2012; Hoang et al., 2020) (Fig. 1a, Fig. S2a). One week following the final tamoxifen dose, we performed intravitreal injections of either Gfap-*Oct4*-mCherry or Gfap-mCherry control 7m8 AAV (Fig. 1b, Fig. S2b). While expression of other neurogenic bHLH factors driven by the GFAP promoter were rapidly silenced in Müller glia and activated in amacrine and retinal ganglion cells, Gfap-*Oct4*-mCherry remained selectively expressed in Müller glia. However, *Oct4* expression alone did not induce detectable levels of Müller glia-derived neurogenesis in the uninjured retina (Le et al., 2022).

Since injury is often required to promote Müller glia-derived neurogenesis following *Ascl1* overexpression or loss of function of *Nfia/b/x* (Hoang et al., 2020; Jorstad et al., 2017), we hypothesized that this might also be so for Müller glia overexpressing *Oct4*. To test this, we injected N-methyl-D-aspartate (NMDA) one week post-viral injection. We then administered 5-ethynyl-2'-deoxyuridine (EdU) to label proliferating cells. Retinas were harvested and analyzed at 1, 2, and 5 weeks post-injury (Fig. S2b, Fig. 1b). Immunostaining analysis revealed enriched mCherry expression in *GlastCreER;Sun1-GFP* Müller glia in both control mCherry and *Oct4*-mCherry infected retinas at all timepoints (Fig. 1c, Fig. S2c-l). In addition, *Oct4* expression was

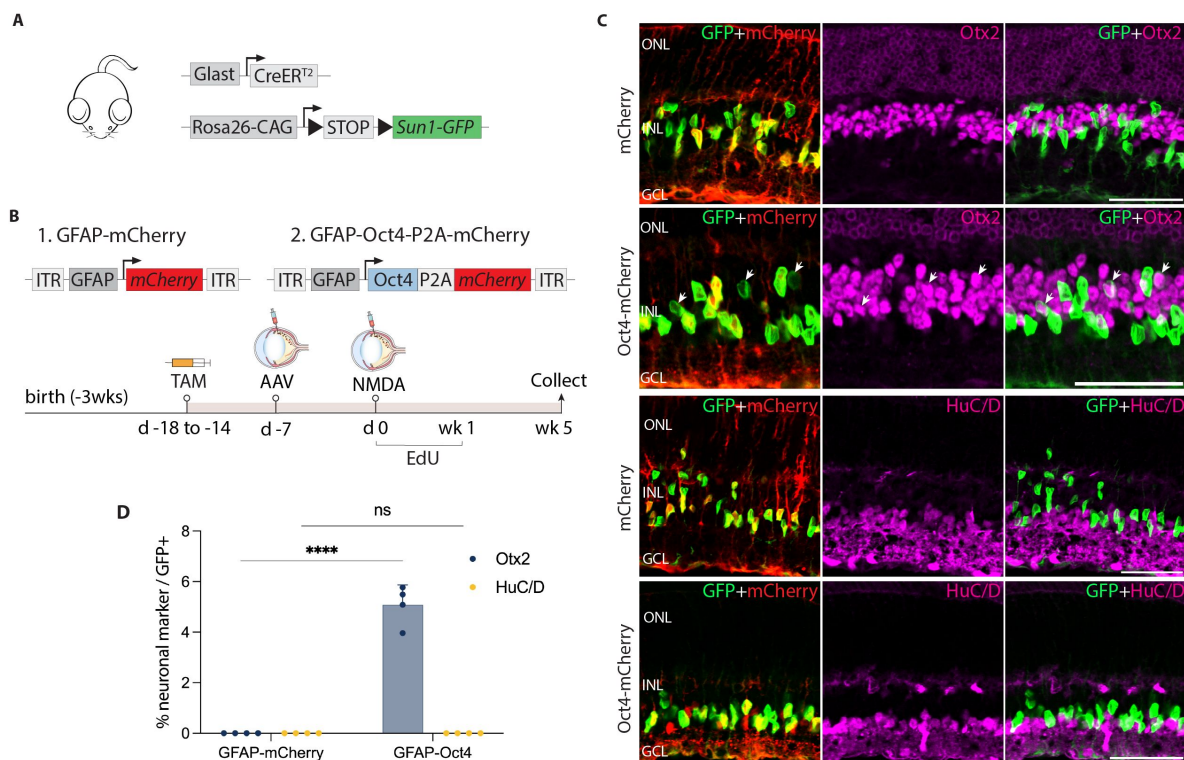


Figure 1

AAV-mediated overexpression of *Oct4* induces neurogenesis in control *GlastrCreER;Sun1-GFP* Müller glia following NMDA-induced excitotoxicity.

(A) Schematic of the transgenic construct used to specifically label Müller glia with Sun1-GFP expression. **(B)** Schematic of the GFAP AAV constructs and experimental workflow. **(C)** Representative images of retinas immunolabeled for GFP, mCherry, Otx2, and HuC/D. White arrowheads indicate GFP-positive Müller glia-derived neurons expressing neuronal markers Otx2 or HuC/D. **(D)** Quantification of mean percentage $\pm$ SD of GFP-positive Müller glia-derived neurons expressing either Otx2 or HuC/D. Significance was determined via two-way ANOVA with Tukey's multiple comparison test: ****p < 0.0001. Each data point was calculated from an individual retina. TAM, tamoxifen; ONL, outer nuclear layer; INL, inner nuclear layer; GCL, ganglion cell layer. Scale bar = 50 μ m.

detected in GFP-positive Müller glia in Oct4-mCherry infected retinas but not in mCherry infected *GlastCreER;Sun1-GFP* retina (Fig. S2c,d, Fig. S3a). While most Sun1-GFP-positive cells infected with Oct4-mCherry expressed the Müller glial marker Sox2 (Fig. S2e,f, S3b), a subset of GFP-positive cells lost Sox2 expression from 2 weeks post-injury onward (Fig. S2f, Fig. S3b, yellow arrows), which is consistent with glia-to-neuron conversion.

We observed selective expression of the neurogenic bHLH factor Ascl1 in Oct4-mCherry infected GFP-positive Müller glia at 1 week post-injury (Fig. S2g), with the number of Ascl1-positive cells declining somewhat by 2 and 5 weeks post-injury (Fig. S2h, Fig. S3c). While at 1 week post-injury, we did not observe any significant Müller glia-derived neurogenesis (Fig. S2i,m), by 2 weeks post-injury we observed 1.1% of GFP-positive cells co-localized with the bipolar cell marker Otx2 in Oct4-mCherry infected retinas (Fig. S2j,m), and this increased to 5.1% of GFP-positive cells at 5 weeks post-injury (Fig. 1c,d). We found that Müller glia-derived Otx2+ bipolar-like cells lost mCherry reporter expression, likely due to reduced activity of the Gfap-Oct4-mCherry construct in neurons (Le et al., 2022). Immunostaining with additional cell-specific markers revealed that a subset of these GFP-positive cells expressed the cone bipolar markers Scgn and Cabp5 (Fig. S3d,e), but not the rod and cone ON bipolar cell marker Isl1 or the rod marker Nrl (Fig. S3f,g). Furthermore, we did not observe expression of the amacrine cell marker HuC/D in GFP-positive cells (Fig. 1c,d), which was observed in neurons generated from neurogenic mouse Müller glia deficient in either *Rbpj* or *Nfia/b/x* (Hoang et al., 2020; Le et al., 2024). No EdU/GFP-positive cells were detected at either 1 week (Fig. S2k) or 5 weeks post-injury (Fig. S3h). However, at 2 weeks post-injury, we observed a very limited number of GFP-positive cells incorporating EdU in retinas infected with both mCherry control and Oct4-mCherry, with modest EdU incorporation but significantly increased by *Oct4* overexpression (Fig. S2l,n).

These findings indicate that *Oct4* overexpression induces a very low level of proliferation in Müller glia in the second week following NMDA injury. However, the absence of EdU/Otx2/GFP-positive cells at both 2 and 5 weeks post-injury suggests that bipolar cell generation at this stage was, at least in part, driven by direct transdifferentiation.

Oct4 overexpression synergistically enhances glial-derived neurogenesis induced by genetic disruption of Notch signaling

In our previous study, we demonstrated that disrupting Notch signaling in adult mouse Müller glia – either by selective deletion of the transcriptional mediator *Rbpj* or by combined disruption of *Notch1* and *Notch2* receptors – induces transdifferentiation into bipolar and amacrine-like cells (Le et al., 2024). To further determine the effects of *Oct4* overexpression in the absence of Notch signaling, we used *GlastCreER;Rbpj^{lox/lox};Sun1-GFP* mice to genetically disrupt *Rbpj* function (Fig. 2a, Fig. S4a). We then performed intravitreal injection of either control or *Oct4* overexpression constructs followed by NMDA and EdU injection as performed for *GlastCreER;Sun1-GFP* mice. Retinas were collected at 1, 2, and 5 weeks following NMDA treatment (Fig. 2b, Fig. S4b). Immunostaining confirmed expression of Oct4 protein in GFP-positive Müller glia of Oct4-mCherry infected retinas, but not in age-matched mCherry-infected controls (Fig. S4c,d, Fig. S5a). We observed that many GFP-positive cells in Oct4-mCherry-infected retinas did not express the Müller glial markers Sox2 (Fig. S4e,f) or Sox9 (Fig. S5b, yellow arrows), suggesting that they have lost their glial identity.

At 1 and 2 weeks post retinal injury, we observed selective expression of the neurogenic bHLH factor Ascl1 in both control mCherry and Oct4-mCherry infected GFP-positive *Rbpj*-deficient Müller glia (Fig. S4g,h). Time course analysis showed that Müller glia-derived cells expressing the bipolar cell marker Otx2 appeared as early as 1 week following NMDA injury (Fig. S4i-k). However, we did not observe a significant change in the proportion of GFP-positive *Rbpj*-deficient Müller

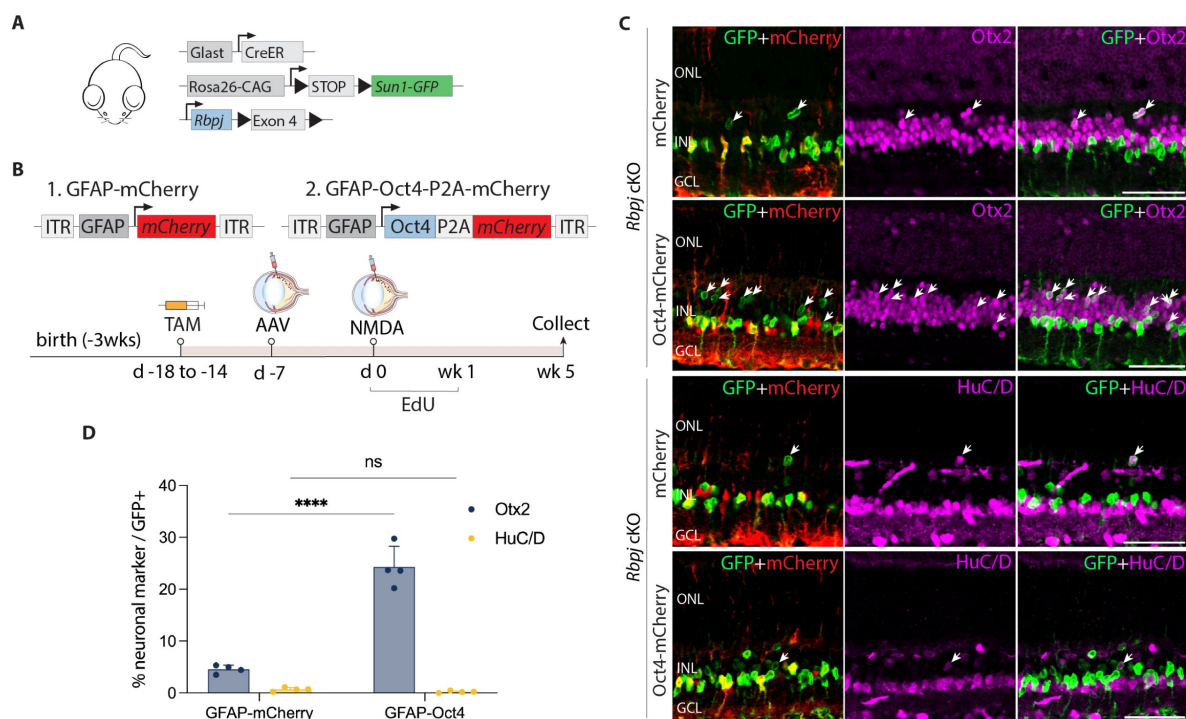


Figure 2

***Oct4* overexpression enhances neurogenesis in *Rbpj*-deficient *GlaxtCreER;Rbpj^{lox/lox};Sun1-GFP* Müller glia.**

(A) Schematic of the transgenic constructs used to induce loss of function of *Rbpj* specifically in Müller glia. (B) Schematic of the GFAP AAV constructs and experimental workflow. (C) Representative images of retinas immunolabeled for GFP, mCherry, Otx2, and HuC/D. White arrowheads indicate GFP-positive Müller glia-derived neurons expressing neuronal markers Otx2 or HuC/D. (D) Quantification of mean percentage $\pm$ SD of GFP-positive Müller glia-derived neurons expressing either Otx2 or HuC/D. Significance was determined via two-way ANOVA with Tukey's multiple comparison test: **** $p < 0.0001$. TAM, tamoxifen; ONL, outer nuclear layer; INL, inner nuclear layer; GCL, ganglion cell layer. Scale bar = 50 μ m.

glia expressing the amacrine cell marker HuC/D. By five weeks after NMDA injury, we observed 4.5% of GFP-positive cells in retinas of mCherry-infected controls expressed Otx2, while 0.7% expressed HuC/D in Oct4-mCherry infected retinas (**Fig. 2c,d**).

These levels of neurogenesis are consistent with previous findings in uninfected *Rbpj*-deficient Müller glia (Le et al., 2024). In contrast, we observed 24.3% of GFP-positive cells expressing Otx2 in Oct4-mCherry-infected retinas, representing a significant ($p < 0.0001$) increase relative to mCherry control. Unlike in control (*GlastCreER;Sun1-GFP*) retinas, Oct4-expressing, *Rbpj*-deficient Müller glia showed no EdU incorporation at either 1, 2 or 5 weeks following infection (Fig. S4l,m, Fig. S5c). EdU labeling instead colocalized with the microglia marker Iba1. The absence of Müller glia proliferation indicates that the observed neurogenesis is due largely to direct transdifferentiation of Müller glia into retinal neuron-like cells, rather than through dedifferentiation to a proliferative neuronal progenitor-like state.

We have previously shown that Müller glia-specific *Notch1/2* double loss of function mutants phenocopy neurogenesis seen in *Rbpj* mutants (Le et al., 2024). We then examined the effect of Oct4 overexpression on *GlastCreER;Notch1^{lox/lox};Notch2^{lox/lox};Sun1-GFP* mice using the same protocol previously described for control and *Rbpj*-deficient Müller glia (Fig. S6a,b). We observed 15.8% of GFP+ Müller glia colocalized with Otx2 in Oct4-mCherry-infected retinas compared to 6.6% GFP/Otx2+ cells in samples infected with control mCherry vector (Fig. S6c,d). As expected, infection with Oct4-mCherry vector induced both Oct4 (Fig. S6e) and Ascl1 (Fig. S6f) expression in *Notch1/2*-deficient Müller glia.

Finally, we tested whether Oct4 overexpression could enhance Müller glia-derived neurogenesis induced by loss of function of the transcription factors *Nfia*, *Nfib*, and *Nfix* (Hoang et al., 2020). To test this, we injected *GlastCreER;Nfia^{lox/lox};Nfib^{lox/lox};Nfix^{lox/lox};Sun1-GFP* mice with both mCherry and Oct4-mCherry vectors as previously described, then analyzed the retinas 5 weeks after NMDA injury (Fig. S7a,b). We observed limited coexpression of mCherry and GFP following infection with either construct (Fig. S7c), indicating that NFI factors may be required for appropriate glial-specific expression of the GFAP minipromoter construct, as suggested by previous findings in astrocytes (Cebolla and Vallejo, 2006).

Oct4 overexpression promotes expression of reactive glial markers and neurogenic bHLH factors in *Rbpj*-deficient Müller glia

To gain insight into the mechanism by which Oct4 stimulates neurogenesis in *Rbpj*-deficient Müller glia, we conducted single-cell (sc)RNA-Seq analysis of GFP-positive cells FACS-isolated from retinas of *GlastCreER;Rbpj^{lox/lox};Sun1-GFP* mice infected with either GFAP-mCherry control or GFAP-Oct4-mCherry virus. Mice were tamoxifen- and NMDA-treated as described in **Figure 2**, and GFP-positive cells were isolated at 8 weeks following NMDA treatment (**Fig. 3a**). We profiled 12,143 GFP+ cells from control-infected and 14,612 GFP+ cells from Oct4-mCherry-infected retinas (**Fig. 3b**).

In both samples, we observed a clear differentiation trajectory connecting Müller glia, neurogenic Müller glia-derived progenitor cells (MGPCs), and differentiating amacrine and bipolar cells (**Fig. 3b**). Oct4-overexpressing samples consistently show substantially higher fractions of bipolar cells (20.5 % vs. 2.7 %), and a smaller fraction of Müller glia (71.4 % vs. 88.2 %), consistent with immunohistochemical data (**Fig. 3c**). Small numbers of contaminating rod and cone photoreceptors were also observed, as previously reported (Hoang et al., 2020), but no immature photoreceptor-like cells were observed, indicating that these represent contaminating mature cells that were not excluded by FACS analysis.

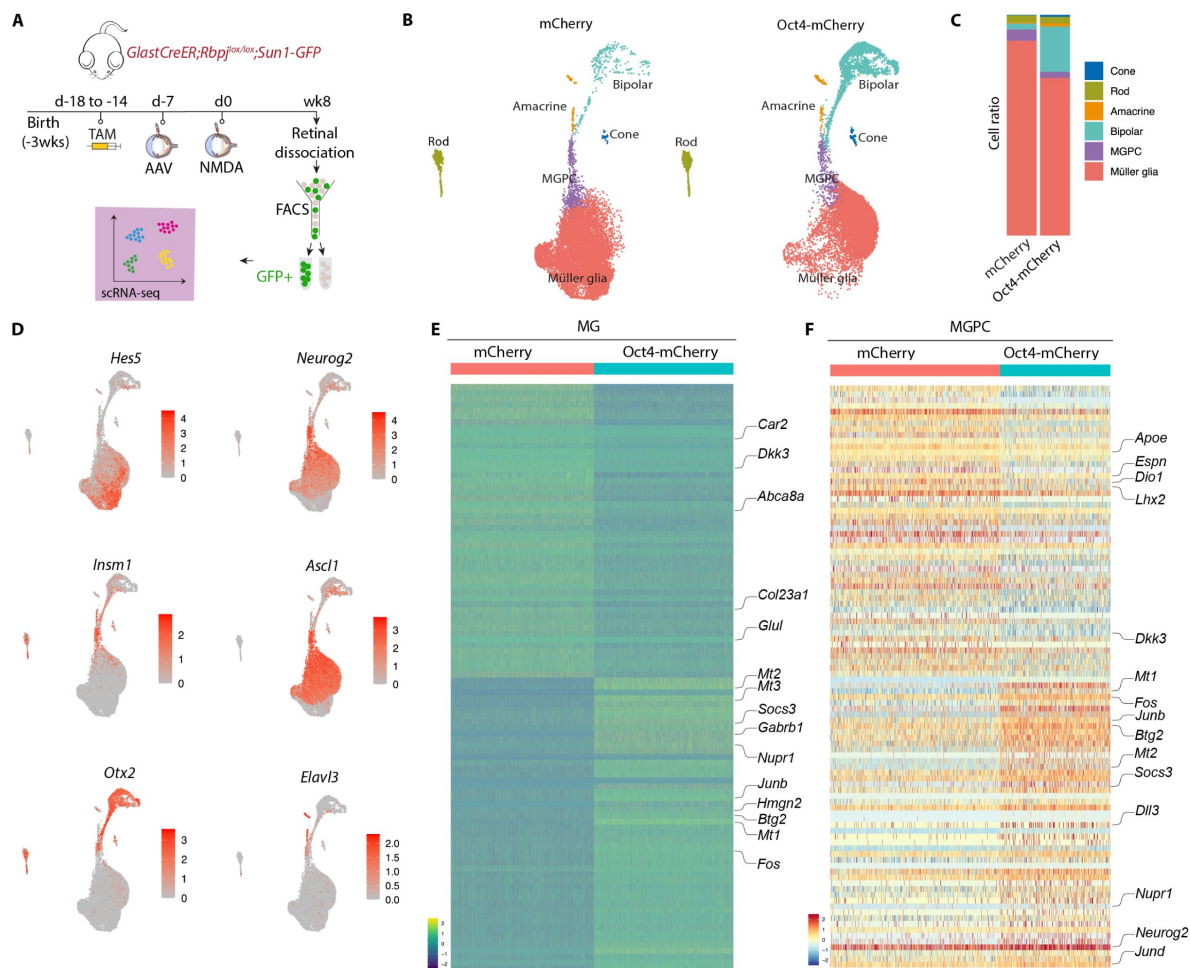


Figure 3

scRNA-Seq analysis of Müller glia and Müller glia-derived neurons from *Rbpj*-deficient *GlastCreER;Sun1-GFP;Rbpj^{lox/lox}* Müller glia following *Oct4* overexpression.

(A) Schematic of the scRNA-seq experimental pipeline. **(B)** UMAP plot showing the clustering of GFP-positive cells from *Rbpj*-deficient *GlastCreER;Rbpj^{lox/lox};Sun1-GFP* Müller glia retinas infected with GFAP-mCherry and GFAP-Oct4-mCherry AAV constructs. **(C)** Stacked bar plots showing the proportion of cells in each cluster across two sample groups. **(D)** Feature plots highlighting the cluster of Müller glia (*Hes5*), neurogenic MGPCs (*Neurog2*, *Insm1*, *Ascl1*), bipolar cells (*Otx2*), and amacrine cells (*Elavl3*). **(E)** Heatmap showing the expression of top differentially expressed genes (DEGs) from the *Rbpj*-deficient *GlastCreER;Rbpj^{lox/lox};Sun1-GFP* Müller glia cell cluster from retinas infected with GFAP-mCherry and GFAP-Oct4-mCherry AAV constructs. **(F)** Heatmap showing the expression of top differentially expressed genes (DEGs) for MGPC cell cluster from retinas infected with GFAP-mCherry and GFAP-Oct4-mCherry AAV constructs.

Selective molecular markers could readily distinguish Müller glia (*Glul*, *Apoe*, *Vim*, *Clu*, *Rbpl1*) from neurogenic MGPCs (*Neurog2*, *Ascl1*, *Sox4*, *Hes6*, *Dll1*), and clearly identify differentiating bipolar (*Otx2*, *Pcp2*, *Pcp4*, *Car10*, *Gabrb3*) and amacrine-like cells (*Nrxn1*, *Tfap2b*, *Nrg1*, *Snap25*) (Fig. 3d-f, Fig. S8a-c, Table ST1). In both MGs and MGPCs, we observed substantial differences in gene expression between samples infected with control and Oct4-mCherry samples. Oct4-infected Müller glia selectively downregulated genes specific to resting Müller glia (*Car2*, *Dkk3*, *Glul*, *Col23a1*, *Abca8a*), and upregulated genes enriched in reactive glia (*Gfap*, *Socs3*, *Nupr1*, *Mt1/2/3*, *Fos*, *Jun*, *Klf6*) (Fig. 3e-f, Fig. 8b-c, Table ST1). In MGPCs, Oct4 overexpression also downregulated both transcription factors (*Lhx2*, *Hopx*) and other genes (*Apoe*, *Dio1*, *Il33*, *Espn*) that are enriched in resting Müller glia, and upregulated many other genes that are selectively expressed in activated glia (*Mt1/2/3*, *Fos*, *Jun*, *Nupr1*, *Socs3*). In addition, Oct4-infected MGPCs expressed consistently higher levels of neurogenic bHLH factors (*Ascl1*, *Neurog2*, *Hes6*), bipolar cell specification (*Otx2*), or genes broadly associated with neurogenesis (*Insm1*, *Dll3*, *Mybl1*). Oct4-infected MGPCs also expressed genes (*Gadd45a*, *Btg2*) enriched in late-stage neurogenic progenitors (Clark et al., 2019) (Fig. 3e-f, Table ST1). Together, these findings imply that Oct4 enhances bipolar cell formation in *Rbpj*-deficient Müller glia potentially by repressing expression of genes such as *Lhx2* that promote quiescence (de Melo et al., 2018, 2016), increasing expression of genes that broadly promote neurogenesis, and selectively increasing expression of the bipolar cell-promoting factor *Otx2* (Chan et al., 2020; Wang et al., 2014).

Single-cell multiomic analysis identifies targets of Oct4 in control and *Rbpj*-deficient Müller glia

To further investigate the mechanism by which Oct4 promotes neurogenesis, we conducted simultaneous single-nucleus (sn)RNA-Seq and ATAC-Seq on GFP-positive cells from both control and *Rbpj*-deficient Müller glia infected with either control or Oct4-mCherry vectors at 8 weeks following NMDA injection (Fig. 4a). UMAP analysis of integrated snRNA/ATAC-Seq profiles was then used to identify cell clusters. Müller glia from control and *Rbpj*-deficient animals form two separate Müller glia clusters (Fig. 4b,c). Control Oct4-expressing Müller glia gave rise to both small numbers of neurogenic MGPCs (Fig. 4d), in line with the small numbers of *Ascl1*-positive cells observed (Fig. S2g,h, Fig. S3b), as well as mature bipolar cells, consistent with histological findings (Fig. 1c,d). Oct4 overexpression substantially enhances bipolar cell generation from *Rbpj*-deficient MGPC (Fig. 4d, Fig. S9a-c). Small numbers of rod and cone photoreceptors were also detected in this analysis, but their proportions did not substantially change between treatment conditions and clearly defined immature transitional states were identified, and therefore likely represent contaminating native cells.

Oct4 overexpression leads to significant changes in gene expression in both control and *Rbpj*-deficient Müller glia. The transition from resting to neurogenic MGPCs to bipolar cells is associated with downregulation of genes that are specifically expressed in resting Müller glia, including *Aqp4*, *Hes1*, *Tcf7l2*, and *Lhx2* (Fig. 4e,f, Fig. S9a-c, Table ST2). Similarly, motif analysis showed that chromatin accessibility is reduced for target motifs for transcription factors that maintain glial quiescence such as *Nfia/b/x* and *Lhx2* (Fig. 4g, Table ST2). MGPCs likewise show increased accessibility at *Stat1/3* motifs, as well as target motifs for neurogenic transcription factors such as *Ascl1*, *Neurog2*, and *Neurod1* (Fig. 4g), consistent with previous studies of MGPCs generated from *Nfia/b/x* and *Rbpj*-deficient Müller glia (Hoang et al., 2020; Le et al., 2024).

Oct4 overexpression in both control and *Rbpj*-deficient Müller glia clusters respectively induces upregulation of 534 and 1657 genes, while respectively downregulating 439 and 824 genes (Fig. 4h). A total of 156 genes are upregulated in both control and *Rbpj*-deficient Müller glia, while 63 genes are downregulated. In both control and *Rbpj*-deficient Müller glia, Oct4 upregulates multiple classes of ion channels, including mechanosensitive (*Piezo2*), TRP (*Trpm1*, *Trpm3*), and chloride channels (*Clca3a1*, *Clca3a2*), as well as GABA receptors (*Gabra4*, *Gabrb1*), cyclic nucleotide phosphodiesterases (*Pde8b*, *Pde11a*), and the neuropeptide *Npy*. Oct4 overexpression also

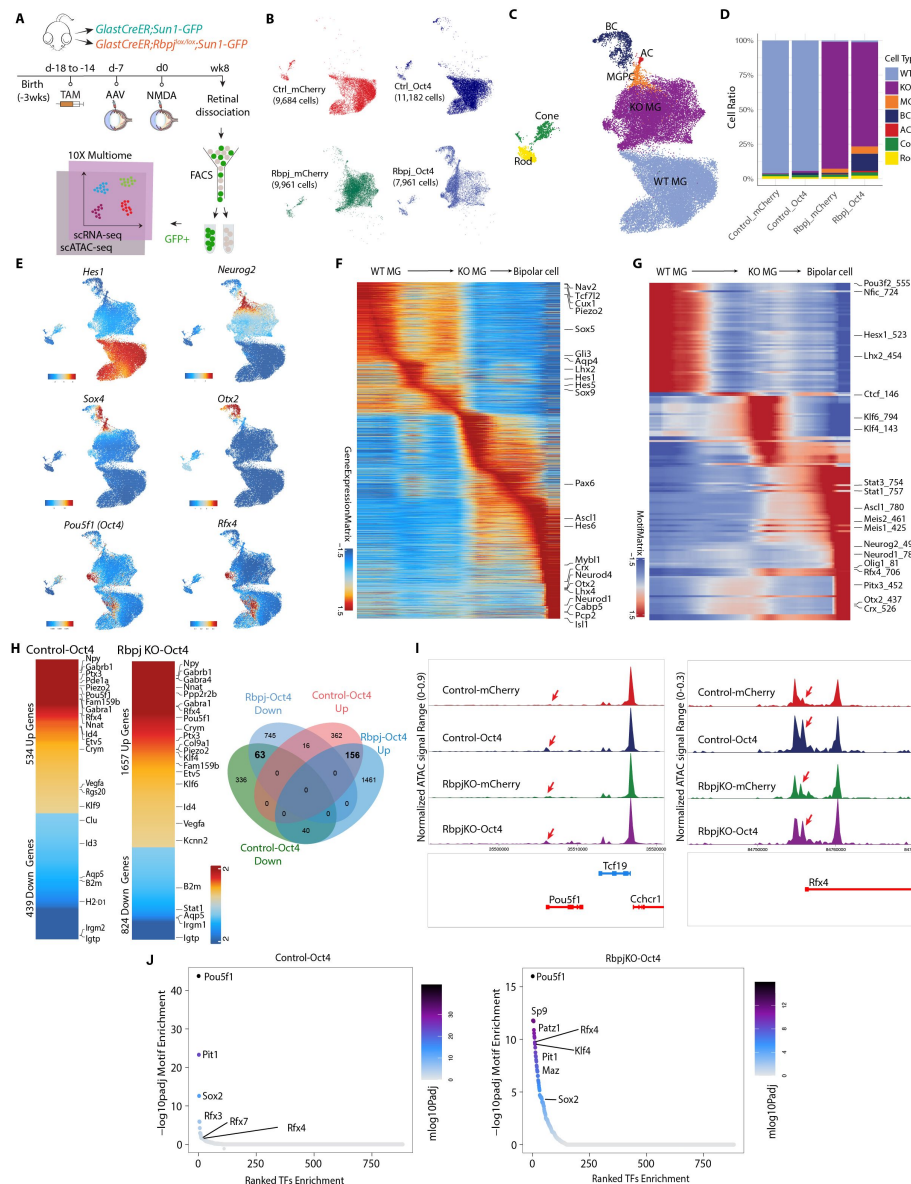


Figure 4

Integrated snRNA/scATAC-Seq analysis of control *GlastCreER;Sun1-GFP* and *Rbpj*-deficient *GlastCreER;Rbpj^{lox/lox};Sun1-GFP* Müller glia following Oct4 overexpression.

(A) Schematic of the multiomic scRNA/ATAC-seq experimental pipeline. **(B)** UMAP plot of multiomic datasets showing the clustering of GFP+ cells from control *GlastCreER;Sun1-GFP* and *Rbpj*-deficient *GlastCreER;Rbpj^{lox/lox};Sun1-GFP* Müller glia from GFAP-mCherry and GFAP-Oct4-mCherry AAV infected retinas. **(C)** UMAP plot showing the identity of cell clusters determined by marker gene expression. **(D)** Stacked bar plots represent the proportion of cells in each cluster across different sample groups. **(E)** Feature plots highlighting the cluster of Müller glia (*Hes1*), neurogenic MGPCs (*Neurog2*, *Sox4*), bipolar cells (*Otx2*), *Pou5f1*- and *Rfx4*-expressing cells. **(F)** Heatmap showing expression of DEGs along the neurogenesis trajectory. **(G)** Heatmap showing differential motif activity along the neurogenesis trajectory. **(H)** Heatmaps of DEGs differentially expressed between Oct4 and mCherry control samples in the control *GlastCreER;Sun1-GFP* and *Rbpj*-deficient *GlastCreER;Rbpj^{lox/lox};Sun1-GFP* Müller glia clusters. Venn diagram showing unique and common DEGs between Oct4 and mCherry control in the control *GlastCreER;Sun1-GFP* and *Rbpj*-deficient *GlastCreER;Rbpj^{lox/lox};Sun1-GFP* Müller glia. **(I)** Increased chromatin accessibility regions associated with the Oct4 and *Rfx4* loci observed following Oct4 overexpression in both control *GlastCreER;Sun1-GFP* and *Rbpj*-deficient *GlastCreER;Rbpj^{lox/lox};Sun1-GFP* Müller glia. **(J)** Top-ranked enriched motifs in chromatin regions showing increased accessibility following Oct4 overexpression in both control *GlastCreER;Sun1-GFP* and *Rbpj*-deficient *GlastCreER;Sun1-GFP;Rbpj^{lox/lox}* Müller glia cell clusters.

selectively downregulates multiple inflammation-associated genes (*B2m*, *H2-D1*), small GTPases (*Gbp2*, *Igtp*, *Irgm2*, *Rab13*, *Rgs20*, *Tgtp2*), along with the glycolytic enzyme *Gapdh* and many ribosomal subunits (Fig. 4h, Table ST3). This comprises a functionally diverse range of genes that are often not prominently expressed in retinal progenitors, and whose role in controlling neurogenic competence remains to be investigated.

Oct4 also differentially regulates multiple transcription factors in both control and *Rbpj*-deficient Müller glia. As expected, *Pou5f1* mRNA is strongly upregulated (Fig. 4e,h), with increased chromatin accessibility also being observed at the endogenous *Pou5f1* promoter, indicating that Oct4 regulates its self-expression (Fig. 4i, Table ST4). Similarly, *Oct4* overexpression strongly upregulates the transcription factor *Rfx4* and accessibility at its endogenous promoter (Fig. 4e,h,i, Table ST4). Previous studies showed that overexpression of *Rfx4* is sufficient to induce neurogenesis when overexpressed in human embryonic stem cells (Choi et al., 2024; Joung et al., 2024). However, *Rfx4* expression is not detectably expressed in developing mouse retinas (Clark et al., 2019), indicating that Oct4-induced neurogenesis may involve mechanisms beyond simply inducing dedifferentiation to a retinal progenitor-like state.

Several other transcription factors are only differentially regulated following *Oct4* overexpression in *Rbpj*-deficient Müller glia. One notable upregulated transcription factor is *Klf4*, which cooperates with Oct4 to induce pluripotency (Takahashi and Yamanaka, 2006). In contrast, *Stat1*, which inhibits neurogenesis induced by *Ascl1* overexpression in Müller glia (Jorstad et al., 2020), is selectively downregulated in *Rbpj*-deficient Müller glia (Table ST3).

Analysis of changes in chromatin accessibility induced by Oct4 reveals that the consensus motif for Oct4 and its paralogue Pit1 (*Pou1f1*) show the strongest increase in accessibility, followed by Rfx family motifs (Fig. 4j, Table ST5). Both control and *Rbpj*-deficient Müller glia show increased accessibility associated with Sox2, which directly interacts with Oct4 and cooperates to induce pluripotency (Rizzino, 2013). While Oct4 does not significantly upregulate the pluripotency factor *Klf4* in control Müller glia, it robustly induces *Klf4* expression in *Rbpj*-deficient Müller glia, and likewise selectively induces accessibility at *Klf4* consensus motif sites (Fig. 4j). In addition, *Rbpj*-deficient Müller glia also show smaller but still significant increases in accessibility at target sites for the proneural factor *Neurog2* and the neurogenic factor *Insm1* (Lyu et al., 2021), identifying additional mechanisms by which Oct4 might synergistically enhance neurogenesis (Fig. 4j, Table ST5).

Discussion

In this study, we demonstrate that AAV-mediated overexpression of Oct4 is sufficient to induce neurogenic competence in Müller glia in adult mice, and to synergistically enhance neurogenesis induced by genetic disruption of Notch signaling. While Oct4 overexpression induces very low levels of proliferation, Oct4-induced neurogenesis appears to mostly result from direct glia-to-bipolar cell transdifferentiation by simultaneously upregulating expression of neurogenic bHLH factors and the bipolar cell promoting factor *Otx2*. In addition, Oct4 upregulates expression of the neurogenic transcription factor *Rfx4*, which is not expressed during retinal development. Oct4 exerts reprogramming effects by broadly increasing chromatin accessibility at predicted regulatory sites associated with these genes.

While we did not detect endogenous *Oct4* expression in either injured or neurogenic mouse Müller glia, the fact that these cells express robust levels of *Sox2*, *Klf4*, and *Myc* implies that they may already be prime to reprogramming induced by Oct4. Our findings differ from two previous reports that initially motivated us to carry out this study. A previous study in zebrafish showed weak levels of the Oct4 homolog *pou5f3* in resting glia, with these levels increasing dramatically following injury (Sharma et al., 2019). It remains unclear why we were unable to detect more

than trace levels of *pou5f3* expression in zebrafish Müller glia in any of the many bulk and single-cell RNA-Seq samples we analyzed (Hoang et al., 2020). We also did not detect the previously reported low levels of injury-induced *Oct4* expression in mouse Müller glia (Reyes-Aguirre and Lamas, 2016). Similar discrepancies in observed cellular patterns of *Oct4* expression have been observed in many other tissues (Lengner et al., 2008), and further work will be needed to clarify these discrepancies.

This study represents a clear instance in which viral-mediated expression coupled with genetic cell lineage analysis was effectively used to induce the conversion of adult mammalian Müller glia into retinal neurons *in situ*. A similar approach has been also successfully used to generate bipolar neurons from both immature (Pollak et al., 2013) and very recently also in mature, Müller glia by overexpression of the neurogenic bHLH factors *Ascl1* and *Atoh1* (Pavlou et al., 2024). Other studies reported efficient conversion of Müller glia to photoreceptors using a complex cocktail of overexpressed factors, including *Ctnnb1*, *Otx2*, *Crx*, and *Nrl* (Yao et al., 2018), and efficient conversion of Müller glia to retinal ganglion cells following overexpression of *Atoh7* and *Pou4f2* (Xiao et al., 2021). However, these studies both used GFAP minipromoters to both drive expression of these constructs and to infer cell lineage. Multiple studies have shown that GFAP minipromoter specificity is heavily influenced by insert sequences present in AAVs, which can result in both silencing of expression in glia and ectopic activation of expression in neurons (Le et al., 2022; L.-L. Wang et al., 2021). As a result, it cannot be concluded that these represent bona fide glia-to-neuron conversion *in vivo*. Similar claims have been made about knockdown of the RNA binding protein *Pthp1*, which was variously claimed to directly convert Müller glia to either retinal ganglion cells or photoreceptors, but has been debunked using genetic cell lineage analysis (Hoang et al., 2022; Xie et al., 2022). In this study, the combination of prospective cell lineage analysis using the *GlastCreER;Sun1-GFP* transgene before infection with GFAP-*Oct4*-mCherry vector, in combination with scRNA-Seq analysis of infected cells, rules out any potentially confounding factors resulting from the GFAP minipromoter.

Our finding that *Oct4* can synergistically enhance neurogenesis induced by genetic disruption of Notch signaling raises the question of whether this might be more broadly applicable to other models of induced neurogenesis in mammalian Müller glia. The mechanism by which *Oct4* induces glial-derived neurogenesis appears complex, and to differ fundamentally from that seen following either overexpression of *Ascl1* or loss of function of *Nfia/b/x* and/or *Rbpj* (Hoang et al., 2020; Jorstad et al., 2017; Le et al., 2024). In these cases, Müller glia broadly upregulate proneural genes and/or downregulate Notch signaling. *Oct4* instead induces expression of the neurogenic transcription factor *Rfx4*, which is not expressed in developing retina. While this may represent a parallel pathway to neurogenic competence that in part accounts for synergistic induction of neurogenesis seen in *Rbpj*-deficient Müller glia, confirmation of the neurogenic activity of *Rfx4* in Müller glia requires direct experimental confirmation.

Other mechanisms enhancing *Oct4*-dependent neurogenesis in the *Rbpj*-deficient Müller glia may include enhanced induction of *Klf4* expression and increased accessibility at target sites for neurogenic transcription factors such as *Ascl1*. While *Ascl1* overexpression induces Müller glia-derived neurogenesis in adult mice, this occurs only following treatment with HDAC inhibitors (Jorstad et al., 2017), even though *Ascl1* is itself a pioneer factor (Păun et al., 2023). *Oct4* overexpression broadly increases the accessibility of putative cis-regulatory sequences associated with both *Neurog2* and *Insm1*, identifying another potential mechanism of synergy.

Transient overexpression of *Oct4* could potentially further enhance the neurogenic function of *Ascl1* and enhance glial-derived neurogenesis more broadly. This may also be the case for pathways reported to induce glial reprogramming in other CNS regions, such as *Sox2* overexpression (Niu et al., 2015). Since *Oct4* overexpression only weakly and transiently induces

proliferation in adult Müller glia, this reduces the potential oncogenic risk, and may ultimately prove to be relevant for the design of cell-based regenerative therapies for treating retinal dystrophies.

Methods

Mice

Mice were raised and housed in a climate-controlled pathogen-free facility on a 14/10 h light/dark cycle. Mice used in this study were *GlastCreER;Sun1-GFP*, which were generated by crossing the *GlastCreER* and *Sun1-GFP* lines developed by Dr. Jeremy Nathans at Johns Hopkins (de Melo et al., 2012 [↗](#); Mo et al., 2015 [↗](#)), and were obtained from his group.

GlastCreER;Rbpj^{lox/lox};Sun1-GFP mice were generated by crossing *GlastCreER;Sun1GFP* with conditional *Rbpj^{lox/lox}* mice (JAX #034200).

GlastCreER;Notch1^{lox/lox};Notch2^{lox/lox};Sun1-GFP mice were obtained by crossing *GlastCreER;Notch1^{lox/lox};Sun1-GFP* and *GlastCreER;Notch2^{lox/lox};Sun1-GFP* mice. *GlastCreER;Nfia/b/x^{lox/lox};Sun1-GFP* mice were generated by crossing *Nfia^{lox/lox}* (see below); *Nfib^{lox/lox}* (Hsu et al., 2011 [↗](#)), and *Nfix^{lox/lox}* (Campbell et al., 2008 [↗](#)) mice to *GlastCreER;Sun1-GFP* mice. *Nfia^{lox/lox}* mice were generated in the Roswell Park Gene Targeting and Transgenic Shared Resource using heterozygous targeted ES cells from EUComm project 38437 (KOMP). All mice used in this study contain both the *GlastCreER* and *Sun1-GFP* transgenes, allowing visualization of both Müller glia and Müller glia-derived neurons. Maintenance and experimental procedures performed on mice were in accordance with the protocol approved by the Institutional Animal Care and Use Committee (IACUC) at the Johns Hopkins School of Medicine.

Intraperitoneal tamoxifen (TAM) Injection

To induce *Cre* recombination, animals at ~3 weeks of age were intraperitoneally injected with tamoxifen (Sigma-Aldrich, #H6278-50mg) in corn oil (Sigma-Aldrich, #C8267-500ML) at 1.5mg/dose for five consecutive days.

NMDA treatment

Adult mice were anesthetized with isoflurane inhalation. Two microliters of 100 mM NMDA in PBS were intravitreally injected using a microsyringe with a 33G blunt-ended needle.

EdU treatment

At 1-week post tamoxifen induction, mice were administered EdU (Abcam, #ab146186) via i.p. injections (150µl of 5mg/ml in PBS) and drinking water (0.3 mg/ml) for 7 consecutive days.

Cloning, production, and intravitreal injection of adeno-associated virus

The Addgene #50473 construct, which contains a GFAP promoter, was used in this study. The EGFP sequence was replaced by the mCherry sequence. The P2A ribosomal self-cleaving peptide is used to simultaneously express *Oct4* 5' to the mCherry reporter as a single polypeptide, which is then cleaved to generate *Oct4* and mCherry. The coding sequence of *Oct4* was synthesized by GeneWiz. AAV constructs were packaged into AAV2.7m8 by Boston Children's Hospital Viral Core. Following tamoxifen induction, one-month-old *GlastCreER;Sun1-GFP* mice were intravitreally injected with GFAP AAV constructs using a microsyringe with a 33G blunt-ended needle. The titres and injection volume for each construct are listed in table ST6.

Immunohistochemistry and imaging

Collection and immunohistochemical analysis of retinas were performed as described previously (Hoang et al., 2020 [DOI](#)). Mouse eye globes were fixed in 4% paraformaldehyde (ElectronMicroscopySciences, #15710) for 4 h at 4°C. Retinas were dissected in 1x PBS and incubated in 30% sucrose overnight at 4°C. Retinas were then embedded in OCT (VWR, #95057-838), cryosectioned at 16 µm thickness, and stored at -20°C. Sections were dried for 30 min in a 37°C incubator and washed 3 × 5 min with 0.1% Triton X-100 in PBS (PBST). EdU labeling was performed by using Click-iT EdU kit (ThermoFisher, #C10340, #C10636) following the manufacturer's instructions. Sections were then incubated in blocking buffer (10% Horse Serum (ThermoFisher, #26050070), 0.4% Triton X-100 in 1x PBS) for 2 h at room temperature (RT) and then incubated with primary antibodies in the blocking buffer overnight at 4°C. Primary antibodies used are listed in table ST7.

Sections were washed 4 × 5 min with PBST to remove excess primary antibodies and were incubated in secondary antibodies in blocking buffer for 2 h at RT. Sections were then counterstained with DAPI in PBST, washed 4 × 5 min in PBST and mounted with ProLong Gold Antifade Mountant (Invitrogen, #P36935) under coverslips (VWR, #48404-453), air-dried, and stored at 4°C. Fluorescent images were captured using a Zeiss LSM 700 confocal microscope. Z-stack images were collected for all sections. Colocalization of markers scored only if observed in individual Z-stack images. Secondary antibodies used are listed in table ST7.

Cell quantification and statistical analysis

Otx2/GFP-positive and HuC/D/GFP-positive cells were counted and divided by the total number of GFP-positive cells from a single random whole section per retina. Each data point in the bar graphs was calculated from an individual retina. All cell quantification data were graphed and analyzed using GraphPad Prism 10. Two-way ANOVA was used for analysis between 3 or more samples of multiple groups. All results are presented as mean ± SD.

Retinal cell dissociation

Retinas were dissected in fresh ice-cold PBS and retinal cells were dissociated using an optimized protocol as previously described (Fadl et al., 2020 [DOI](#)). Each sample contains a minimum of 4 retinas from 4 animals of both sexes. Dissociated cells were resuspended in ice-cold HBAG Buffer containing Hibernate A (BrainBits, #HALF500), B-27 supplement (ThermoFisher, #17504044), and Glutamax (ThermoFisher, #35050061).

Single-cell RNA-sequencing (scRNA-seq) library preparation

ScRNA-seq libraries were prepared using dissociated retinal cells using the 10X Genomics Chromium Single Cell 3' Reagents Kit v3.1 (10X Genomics, Pleasanton, CA). Libraries were constructed following the manufacturer's instructions and were sequenced using Illumina NextSeq. Sequencing data were processed through the Cell Ranger 7.0.1 pipeline (10X Genomics) using default parameters.

Single-cell Multiome ATAC + GEX sequencing library preparation

ScATAC-seq and scRNA-seq libraries were prepared using FACS-isolated GFP-positive cells using the 10X Genomic Chromium Next GEM Single Cell Multiome ATAC + Gene Expression kit following the manufacturer's instructions. Briefly, cells were spun down at 500xg for 5 min, resuspended in 100 µl of ice-cold 0.1x Lysis Buffer, lysed by pipette-mixing 4 times, and incubated on ice for 4 min total. Cells were washed with 0.5 ml of ice-cold Wash Buffer and spun down at 500xg for 5 min at 4°C. Nuclei pellets were resuspended in 10-15 µl Nuclei Buffer and counted using Trypan blue. Resuspended cell nuclei (10-15k) were utilized for transposition and loaded into the 10x Genomics

Chromium Single Cell system. ATAC libraries were amplified with 10 PCR cycles and were sequenced on Illumina NovaSeq with ~200 million reads per library. RNA libraries were amplified from cDNA with 14 PCR cycles and were sequenced on Illumina NovaSeq 6000

scRNA-seq data analysis

ScRNA-Seq data were pre-processed using Cellranger v7.1.0 using a custom mouse genome (version mm10) with mCherry and Sun1GFP sequences included. The cell-by-genes count matrices were further analyzed using Seurat V5.1.0 (Hao et al., 2024 [DOI](#)). Cells with RNA counts less than 1000 or greater than 25,000, and number of genes less than 500 or greater than 6000 were filtered out as low-quality cells. Cells with a mitochondrial fraction of greater than 10% were removed. For data visualization, UMAP was generated using the first 20 dimensions.

Müller glia and MGPC cell clusters were subsetted for further analysis. Differential gene expression of mCherry control and Oct4-mCherry groups was performed using the “FindAllMarkers” function.

Single-cell Multiomic analysis

Raw scRNA-seq and scATAC-seq data were processed with the Cell Ranger software version 2.0.2 specifically using cellranger-arc pipeline for formatting reads, demultiplexing samples, genomic alignment, and generating the cell-by-gene count matrix and the fragments files. Both the cell-by-gene count matrix and the fragments files were the final output from the cellranger-arc pipeline, and were used for all downstream analysis.

ScRNA-seq and scATAC-seq were analyzed using Archr version 1.0.2. EnsDb.Mmusculus.v79 and BSgenome.Mmusculus.UCSC.mm10 were used for annotations. The arrow file for each sample was created using the cell-by-gene count matrix and the fragments files output from the cellranger-arc pipeline. To filter poor quality cells, Gex_nGenes greater than 500 and less than 5000 were kept, Gex_nUMI greater than 1000 and less than 15,000 were kept. TSSEnrichment greater than 10 and nFragments greater than 5000 were kept, while other cells were deemed of poor quality and were filtered out. Gene expression values from paired scATAC-seq and scRNA-seq multi modal assay were added to the arrow files using the “addGeneExpressionMatrix” function. The LSI dimensionality reduction for both scATAC and scRNASeq was computed and combined using the “addCombinedDims” function. UMAP embedding is calculated using “addUMAP” function and clusters are identified from the reduced dimensions. Marker features were used to annotate clusters.

To identify peaks, “addReproduciblePeakSet” function was used to get insertions from coverage files, call peaks, and merge peaks and this step internally used macs software. Then the “addPeakMatrix” function was called to independently compute counts for each peak per cell.

To link peaks to genes, “getPeak2GeneLinks” function was called. The “getMarkerFeatures” function was used to get the differential gene expressions and the differential peaks expressions using the GeneExpressionMatrix and PeakMatrix calculated respectively. Peaks with motifs were annotated using the cisbp motif set using “annotatePeaks.pl” from HOMMER version 4.11 to associate the peaks with nearby genes. The full code used for the analysis is available at <https://github.com/SherineAwad/OCT4scRNA-ATAC> [DOI](#)

Additional Information

Author contributions

N.L. collected and analyzed the bulk of all data presented here. S.A. analyzed single-cell multiomic data, and I.P. assisted in generating multiomic data. T.H. initiated the study, and collected pilot data. S.B. supervised, coordinated, and directed this work. N.L., T.H., and S.B. drafted the manuscript, which all authors edited.

Data and materials availability

All scRNA-seq, snRNA-seq, and scATAC-seq data described in this study are available at Gene Expression Omnibus under accession number GSE277390.

Funding

This study was supported by an award from the National Eye Institute (R01EY031685) to S.B., a Stein Innovation Award from Research to Prevent Blindness to S.B., and a Young Investigator grant from Alcon Research Institute to T.H.

Additional files

Figure S1. *Oct4/pou5f3* mRNA is not detectably expressed in Müller glia after injury. Expression of pluripotency factors in resting or injured retinas by scRNA-Seq and bulk RNA data in (a) zebrafish and (b) mouse after retinal injury from Hoang et al., 2020 and in (c) *Nfia/b/x;Rbpj*-deficient, *Sun1*-GFP-positive Müller glia from Le et al., 2024. [↗](#)

Figure S2. Immunohistochemical analysis of glial markers, AAV reporter expression, and other molecular markers at 1- and 2-weeks following NMDA in control *GlastCreER;Sun1*-GFP Müller glia. (a) Schematic of the transgenic construct used to specifically label Müller glia with *Sun1*-GFP expression. (b) Schematic of the GFAP AAV constructs and experimental workflow. Representative images of retinas immunolabeled for GFP, mCherry and (c-d) *Oct4*, (e-f) *Sox2*, (g-h) *Ascl1*, (i-j) *Otx2*, (k-l) *EdU/Iba1*. White arrowheads indicate GFP-positive cells expressing selected markers. Yellow arrowheads indicate *EdU*-positive cells that do not co-label with *Sox2*, indicating cells that have undergone Müller glia to bipolar cell conversion, and *EdU/Iba1*+ cells that do not co-label with GFP+ cells. Asterisks (*) indicate mouse-on-mouse vascular staining. Quantification of mean percentage $\pm$ SD of GFP-positive Müller glia co-labeled with (m) *Otx2* and (n) *EdU*. Significance was determined via two-way ANOVA with Tukey's multiple comparison test: *** $p < 0.001$, * $p < 0.05$. Each data point was calculated from an individual retina. TAM, tamoxifen; ONL, outer nuclear layer; INL, inner nuclear layer; GCL, ganglion cell layer. Scale bar = 50 μ m. [↗](#)

Figure S3. Immunohistochemical analysis of glial markers, AAV reporter expression, and other molecular markers at 5 weeks following NMDA in control Müller glia. (a-h) Representative images of retinas from control *GlastCreER;Sun1*-GFP mice immunolabeled for GFP, mCherry, and (a) *Oct4*, (b) *Sox2*, (c) *Ascl1*, (d) *Scgn*, (e) *Cabp5*, (f) *Isl1*, (g) *Nrl* and (h) *EdU*. White arrowheads indicate GFP-positive cells co-labeled with the relevant marker. Yellow arrowheads indicate GFP-positive Müller glia that did not co-label with Müller glia markers. Asterisks (*) indicate mouse-on-mouse vascular staining. ONL, outer nuclear layer; INL, inner nuclear layer; GCL, ganglion cell layer. Scale bar = 50 μ m. [↗](#)

Figure S4. Immunohistochemical analysis of glial markers, AAV reporter expression, and other molecular markers at 1- and 2-weeks following NMDA in *Rbpj*-deficient Müller glia. (a) Schematic of the transgenic constructs (*GlastCreER;Rbpj^{lox/lox};Sun1*-GFP) used to induce loss of function of *Rbpj* specifically in Müller glia. (b) Schematic of the GFAP AAV constructs and experimental workflow. Representative images of retinas immunolabeled for GFP, mCherry and (c-d) *Oct4*, (e-f) *Sox2*, (g-h) *Ascl1*, (i-j) *Otx2* and (l-m) *EdU/Iba1*. White arrowheads indicate GFP-positive cells expressing selected markers. Yellow arrowheads indicate GFP-positive cells that do not co-label with *Sox2*, and *EdU/Iba1*+ cells that do not express GFP.

Asterisks (*) indicate mouse-on-mouse vascular staining. **(k)** Quantification of mean percentage $\pm$ SD of GFP-positive Müller glia-derived neurons expressing Otx2. Significance was determined via two-way ANOVA with Tukey's multiple comparison test: * $p < 0.05$. Each data point was calculated from an individual retina. TAM, tamoxifen; ONL, outer nuclear layer; INL, inner nuclear layer; GCL, ganglion cell layer. Scale bar = 50 μ m. [↗](#)

Figure S5. Immunohistochemical analysis of glial markers, AAV reporter expression, and other molecular markers at 5 weeks following NMDA in *Rbpj*-deficient Müller glia. **(a-c)** Representative images of retina from *GlastCreER;Rbpj^{lox/lox};Sun1-GFP* mice immunolabeled for GFP, mCherry, and **(a)** Oct4, **(b)** Sox9, **(c)** EdU. White arrowheads indicate GFP-positive cells co-labeled with the relevant marker. Yellow arrowheads indicate GFP-positive Müller glia that did not co-label with Müller glia markers. Asterisks (*) indicate mouse-on-mouse vascular staining. ONL, outer nuclear layer; INL, inner nuclear layer; GCL, ganglion cell layer. Scale bar = 50 μ m. **(b)** Quantification of mean percentage $\pm$ SD of GFP-positive Müller glia without Sox9 expression. [↗](#)

Figure S6. AAV-mediated overexpression of Oct4 induces neurogenesis in injured Notch1/2-deficient Müller glia. **(a)** Schematic of the transgenic mice (*GlastCreER;Notch1^{lox/lox};Notch2^{lox/lox};Sun1-GFP*) used to induce deletion of Notch1/2 specifically in Müller glia. **(b)** Schematic of the experimental workflow. Representative images of retinas immunolabeled for GFP, mCherry, and **(c)** Otx2, HuC/D, **(e)** Oct4, **(f)** Ascl1. White arrowheads indicate GFP-positive Müller glia expressing Otx2, Oct4, or Ascl1. Asterisks (*) indicate mouse-on-mouse vascular staining. **(d)** Quantification of mean percentage $\pm$ SD of GFP-positive Müller glia-derived neurons expressing either Otx2 or HuC/D. Significance was determined via two-way ANOVA with Tukey's multiple comparison test: *** $p < 0.0001$. TAM, tamoxifen; ONL, outer nuclear layer; INL, inner nuclear layer; GCL, ganglion cell layer. Scale bar = 50 μ m. [↗](#)

Figure S7. Oct4 overexpression does not enhance neurogenesis in *Nfia/b/x*-deficient Müller glia because the GFAP minipromoter is not active in these cells. **(a)** Schematic of the transgenic mice (*GlastCreER;Nfia^{lox/lox};Nfiab^{lox/lox};Nfidx^{lox/lox};Sun1-GFP*) used to induce deletion of *Nfia/b/x* specifically in Müller glia. **(b)** Schematic of the experimental workflow. **(c)** Representative images of retinas immunolabeled for GFP, mCherry and Sox2. White arrowheads indicate mCherry expression in Sox2+ Müller glia that are not Sun1-GFP+. Quantification of mean percentage $\pm$ SD of GFP-positive Müller glia-derived neurons expressing either Otx2 or HuC/D. TAM, tamoxifen; ONL, outer nuclear layer; INL, inner nuclear layer; GCL, ganglion cell layer. Scale bar = 50 μ m. [↗](#)

Figure S8. ScRNA-Seq analysis of Müller glia-derived neurons from *Rbpj*-deficient *GlastCreER;Rbpj^{lox/lox};Sun1-GFP* Müller glia following Oct4 overexpression. UMAP plot showing GFP-positive cells from retinas infected with GFAP-mCherry and GFAP-Oct4-mCherry AAVs. **(b)** Expression of glial and neuronal markers. **(c)** Top DEGs of different cell types generated by Oct4 overexpression in *Rbpj*-deficient Müller glia. [↗](#)

Figure S9. Pseudotime analysis showing the transition from resting to neurogenic MGPCs to bipolar cells. **(a)** Top DEGs across different retinal cell types. **(b)** Pseudotime trajectory of neurogenesis from Müller glia to bipolar cells. **(c)** Heatmap showing differential chromatin accessible regions at different pseudotime stages along the neurogenesis trajectory. [↗](#)

Table ST1. ScRNA-Seq analysis of Sun1-GFP Müller glia-derived cells identifies genes differentially regulated by Oct4 overexpression. Raw+adjusted *p*-values, relative fold-change, and the fraction of all profiled cells expressing the gene in question is shown for both Müller glia and Müller glia-derived progenitor cells transfected with AAV-GFAP-Oct4-mCherry vs. AAV-GFAP-mCherry. [↗](#)

Table ST2. Multiomic analysis identifies DEGs, differential chromatin accessible regions and differentially accessible transcription factor target motifs are shown at discrete pseudotime intervals along the Müller glia-bipolar differentiation trajectory. [↗](#)

Table ST3. SnRNA-Seq analysis identifies DEGs induced by Oct4 overexpression from control *GlastCreER;Sun1-GFP* and *Rbpj*-deficient *GlastCreER;Sun1-GFP;Rbpj^{lox/lox}* Müller glia cell clusters. [↗](#)

Table ST4. SnATAC-Seq analysis identifies differentially accessible chromatin regions regulated by Oct4 overexpression in control *GlastCreER;Sun1-GFP* and *Rbpj*-deficient *GlastCreER;Sun1-GFP;Rbpj^{lox/lox}* Müller glia cell clusters. [↗](#)

Table ST5. SnATAC-Seq analysis identifies differential motif activity in increased accessible chromatin regions by Oct4 overexpression in control *GlastCreER;Sun1-GFP* and *Rbpj*-deficient *GlastCreER;Sun1-GFP;Rbpj^{lox/lox}* Müller glia cell clusters. [↗](#)

Table ST6. List of AAV constructs used in the study. Columns identify the AAV construct, titre, and injection volume. [↗](#)

Table ST7. List of antibodies used in the study. Columns identify the antibody, source, catalog number, and RRID identifier. [↗](#)

References

1. Abad M, Mosteiro L, Pantoja C, Cañamero M, Rayon T, Ors I, Graña O, Megías D, Domínguez O, Martínez D, Manzanares M, Ortega S, Serrano M (2013) **Reprogramming in vivo produces teratomas and iPS cells with totipotency features** *Nature* **502**:340–345
2. Asadi S, Dehghan S, Hajikaram M, Mowla SJ, Ahmadiani AA, Javan M (2015) **Comparing The Effects of Small Molecules BIX-01294, Bay K8644, RG-108 and Valproic Acid, and Their Different Combinations on Induction of Pluripotency Marker-Genes by Oct4 in The Mouse Brain** *Cell J* **16**:416–425
3. Campbell CE, Piper M, Plachez C, Yeh Y-T, Baizer JS, Osinski JM, Litwack ED, Richards LJ, Gronostajski RM (2008) **The transcription factor Nfix is essential for normal brain development** *BMC Dev Biol* **8**:52
4. Cebolla B, Vallejo M (2006) **Nuclear factor-I regulates glial fibrillary acidic protein gene expression in astrocytes differentiated from cortical precursor cells** *J Neurochem* **97**:1057–1070
5. Chan CSY, Lonfat N, Zhao R, Davis AE, Li L, Wu M-R, Lin C-H, Ji Z, Cepko CL, Wang S (2020) **Cell type- and stage-specific expression of Otx2 is regulated by multiple transcription factors and -regulatory modules in the retina** *Development* **147** <https://doi.org/10.1242/dev.187922>
6. Choi W, Choe MS, Kim SM, Kim SJ, Lee J, Lee Y, Lee S-M, Dho SH, Lee M-Y, Kim LK (2024) **RFX4 is an intrinsic factor for neuronal differentiation through induction of proneural genes POU3F2 and NEUROD1** *Cell Mol Life Sci* **81**:99
7. Clark BS, Stein-O'Brien GL, Shiao F, Cannon GH, Davis-Marcisak E, Sherman T, Santiago CP, Hoang TV, Rajaii F, James-Esposito RE, Gronostajski RM, Fertig EJ, Goff LA, Blackshaw S (2019) **Single-Cell RNA-Seq Analysis of Retinal Development Identifies NFI Factors as Regulating Mitotic Exit and Late-Born Cell Specification** *Neuron* **102**:1111–1126
8. Dehghan S, Asadi S, Hajikaram M, Soleimani M, Mowla SJ, Fathollahi Y, Ahmadiani A, Javan M (2015) **Exogenous Oct4 in combination with valproic acid increased neural progenitor markers: an approach for enhancing the repair potential of the brain** *Life Sci* **122**:108–115
9. Dehghan S, Hesaraki M, Soleimani M, Mirnajafi-Zadeh J, Fathollahi Y, Javan M (2016) **Oct4 transcription factor in conjunction with valproic acid accelerates myelin repair in demyelinated optic chiasm in mice** *Neuroscience* **318**:178–189
10. de Melo J, Clark BS, Venkataraman A, Shiao F, Zibetti C, Blackshaw S (2018) **Ldb1- and Rnf12-dependent regulation of Lhx2 controls the relative balance between neurogenesis and gliogenesis in the retina** *Development* **145** <https://doi.org/10.1242/dev.159970>
11. de Melo J, Miki K, Rattner A, Smallwood P, Zibetti C, Hirokawa K, Monuki ES, Campochiaro PA, Blackshaw S (2012) **Injury-independent induction of reactive gliosis in retina by loss of function of the LIM homeodomain transcription factor Lhx2** *Proc Natl Acad Sci U S A* **109**:4657–4662

12. de Melo J, Zibetti C, Clark BS, Hwang W, Miranda-Angulo AL, Qian J, Blackshaw S (2016) **Lhx2 Is an Essential Factor for Retinal Gliogenesis and Notch Signaling** *J Neurosci* **36**:2391–2405
13. Fadl BR, Brodie SA, Malasky M, Boland JF, Kelly MC, Kelley MW, Boger E, Fariss R, Swaroop A, Campello L (2020) **An optimized protocol for retina single-cell RNA sequencing** *Mol Vis* **26**:705–717
14. Gao X, Wang X, Xiong W, Chen J (2016) **In vivo reprogramming reactive glia into iPSCs to produce new neurons in the cortex following traumatic brain injury** *Sci Rep* **6**:22490
15. Hao Y, Stuart T, Kowalski MH, Choudhary S, Hoffman P, Hartman A, Srivastava A, Molla G, Madad S, Fernandez-Granda C, Satija R (2024) **Dictionary learning for integrative, multimodal and scalable single-cell analysis** *Nat Biotechnol* **42**:293–304
16. Hoang T, Kim DW, Appel H, Pannullo NA, Leavey P, Ozawa M, Zheng S, Yu M, Peachey NS, Blackshaw S (2022) **Genetic loss of function of Ptbp1 does not induce glia-to-neuron conversion in retina** *Cell Rep* **39**:110849
17. Hoang T, Wang J, Boyd P, Wang F, Santiago C, Jiang L, Yoo S, Lahne M, Todd LJ, Jia M, Saez C, Keuthan C, Palazzo I, Squires N, Campbell WA, Rajaii F, Parayil T, Trinh V, Kim DW, Wang G, Campbell LJ, Ash J, Fischer AJ, Hyde DR, Qian J, Blackshaw S (2020) **Gene regulatory networks controlling vertebrate retinal regeneration** *Science* **370** <https://doi.org/10.1126/science.abb8598>
18. Hsu Y-C, Osinski J, Campbell CE, Litwack ED, Wang D, Liu S, Bachurski CJ, Gronostajski RM (2011) **Mesenchymal nuclear factor I B regulates cell proliferation and epithelial differentiation during lung maturation** *Dev Biol* **354**:242–252
19. Jorstad NL, Wilken MS, Grimes WN, Wohl SG, VandenBosch LS, Yoshimatsu T, Wong RO, Rieke F, Reh TA (2017) **Stimulation of functional neuronal regeneration from Müller glia in adult mice** *Nature* **548**:103–107
20. Jorstad NL, Wilken MS, Todd L, Finkbeiner C, Nakamura P, Radulovich N, Hooper MJ, Chitsazan A, Wilkerson BA, Rieke F, Reh TA (2020) **STAT Signaling Modifies Ascl1 Chromatin Binding and Limits Neural Regeneration from Muller Glia in Adult Mouse Retina** *Cell Rep* **30**:2195–2208
21. Joung J, Ma S, Tay T, Geiger-Schuller KR, Kirchgatterer PC, Verdine VK, Guo B, Arias-Garcia MA, Allen WE, Singh A, Kuksenko O, Abudayyeh OO, Gootenberg JS, Fu Z, Macrae RK, Buenrostro JD, Regev A, Zhang F (2024) **A transcription factor atlas of directed differentiation** *Cell* **187**:3161
22. Le N, Appel H, Pannullo N, Hoang T, Blackshaw S (2022) **Ectopic insert-dependent neuronal expression of GFAP promoter-driven AAV constructs in adult mouse retina** *Front Cell Dev Biol* **10**:914386
23. Lengner CJ, Welstead GG, Jaenisch R (2008) **The pluripotency regulator Oct4: a role in somatic stem cells?** *Cell Cycle* **7**:725–728
24. Le N, Vu T-D, Palazzo I, Pulya R, Kim Y, Blackshaw S, Hoang T (2024) **Robust reprogramming of glia into neurons by inhibition of Notch signaling and nuclear factor I (NFI) factors in adult mammalian retina** *Sci Adv* **10**:eadn2091
25. Loh Y-H, Wu Q, Chew J-L, Vega VB, Zhang W, Chen X, Bourque G, George J, Leong B, Liu J, Wong K-Y, Sung KW, Lee CWH, Zhao X-D, Chiu K-P, Lipovich L, Kuznetsov VA, Robson P, Stanton LW,

- Wei C-L, Ruan Y, Lim B, Ng H-H (2006) **The Oct4 and Nanog transcription network regulates pluripotency in mouse embryonic stem cells** *Nat Genet* **38**:431–440
26. Lyu P, Hoang T, Santiago CP, Thomas ED, Timms AE, Appel H, Gimmen M, Le N, Jiang L, Kim DW, Chen S, Espinoza DF, Telger AE, Weir K, Clark BS, Cherry TJ, Qian J, Blackshaw S (2021) **Gene regulatory networks controlling temporal patterning, neurogenesis, and cell-fate specification in mammalian retina** *Cell Rep* **37**:109994
27. Mo A, Mukamel EA, Davis FP, Luo C, Henry GL, Picard S, Urich MA, Nery JR, Sejnowski TJ, Lister R, Eddy SR, Ecker JR, Nathans J (2015) **Epigenomic Signatures of Neuronal Diversity in the Mammalian Brain** *Neuron* **86**:1369–1384
28. Morrison GM, Brickman JM (2006) **Conserved roles for Oct4 homologues in maintaining multipotency during early vertebrate development** *Development* **133**:2011–2022
29. Nelson BR, Ueki Y, Reardon S, Karl MO, Georgi S, Hartman BH, Lamba DA, Reh TA (2011) **Genome-wide analysis of Müller glial differentiation reveals a requirement for Notch signaling in postmitotic cells to maintain the glial fate** *PLoS One* **6**:e22817
30. Niu W, Zang T, Smith DK, Vue TY, Zou Y, Bachoo R, Johnson JE, Zhang C-L (2015) **SOX2 reprograms resident astrocytes into neural progenitors in the adult brain** *Stem Cell Reports* **4**:780–794
31. Niu W, Zang T, Zou Y, Fang S, Smith DK, Bachoo R, Zhang C-L (2013) **In vivo reprogramming of astrocytes to neuroblasts in the adult brain** *Nat Cell Biol* **15**:1164–1175
32. Ocampo A, Reddy P, Martinez-Redondo P, Platero-Luengo A, Hatanaka F, Hishida T, Li M, Lam D, Kurita M, Beyret E, Araoka T, Vazquez-Ferrer E, Donoso D, Roman JL, Xu J, Rodriguez Esteban C, Nuñez G, Nuñez Delicado E, Campistol JM, Guillen I, Guillen P, Izpisua Belmonte JC (2016) **In Vivo Amelioration of Age-Associated Hallmarks by Partial Reprogramming** *Cell* **167**:1719–1733
33. Ohnishi K, Semi K, Yamamoto T, Shimizu M, Tanaka A, Mitsunaga K, Okita K, Osafune K, Arioka Y, Maeda T, Soejima H, Moriwaki H, Yamanaka S, Woltjen K, Yamada Y (2014) **Premature termination of reprogramming in vivo leads to cancer development through altered epigenetic regulation** *Cell* **156**:663–677
34. Păun O, Tan YX, Patel H, Strohbuecker S, Ghanate A, Cobolli-Gigli C, Llorian Sopena M, Gerontogianni L, Goldstone R, Ang S-L, Guillemot F, Dias C (2023) **Pioneer factor ASCL1 cooperates with the mSWI/SNF complex at distal regulatory elements to regulate human neural differentiation** *Genes Dev* **37**:218–242
35. Pavlou M, Probst M, Filippova E, Kaplan L, Prieve AR, Rieke F, Reh TA (2024) **AAV-mediated expression of proneural factors stimulates neurogenesis from adult Müller glia in vivo** *bioRxiv* <https://doi.org/10.1101/2024.09.13.612930>
36. Pollak J, Wilken MS, Ueki Y, Cox KE, Sullivan JM, Taylor RJ, Levine EM, Reh TA (2013) **ASCL1 reprograms mouse Muller glia into neurogenic retinal progenitors** *Development* **140**:2619–2631
37. Reyes-Aguirre LI, Lamas M (2016) **Oct4 Methylation-Mediated Silencing As an Epigenetic Barrier Preventing Müller Glia Dedifferentiation in a Murine Model of Retinal Injury** *Front Neurosci* **10**:523

38. Rizzino A (2013) **Concise review: The Sox2-Oct4 connection: critical players in a much larger interdependent network integrated at multiple levels** *Stem Cells* **31**:1033–1039
39. Sharma P, Gupta S, Chaudhary M, Mitra S, Chawla B, Khursheed MA, Ramachandran R (2019) **Oct4 mediates Müller glia reprogramming and cell cycle exit during retina regeneration in zebrafish** *Life Sci Alliance* **2** <https://doi.org/10.26508/lsa.201900548>
40. Sim S-E, Park S-W, Choi S-L, Yu N-K, Ko H-G, Jang D-J, Lee K, Kaang B-K (2011) **Assessment of the effects of virus-mediated limited Oct4 overexpression on the structure of the hippocampus and behavior in mice** *BMB Rep* **44**:793–798
41. Surzenko N, Crowl T, Bachleda A, Langer L, Pevny L (2013) **SOX2 maintains the quiescent progenitor cell state of postnatal retinal Muller glia** *Development* **140**:1445–1456
42. Takahashi K, Yamanaka S (2006) **Induction of pluripotent stem cells from mouse embryonic and adult fibroblast cultures by defined factors** *Cell* **126**:663–676
43. Takashima S, Hirose M, Ogonuki N, Ebisuya M, Inoue K, Kanatsu-Shinohara M, Tanaka T, Nishida E, Ogura A, Shinohara T (2013) **Regulation of pluripotency in male germline stem cells by Dmrt1** *Genes Dev* **27**:1949–1958
44. Uhlen M, Oksvold P, Fagerberg L, Lundberg E, Jonasson K, Forsberg M, Zwahlen M, Kampf C, Wester K, Hober S, Wernerus H, Björling L, Ponten F (2010) **Towards a knowledge-based Human Protein Atlas** *Nat Biotechnol* **28**:1248–1250
45. Wang C, Rabadan Ros R, Martinez-Redondo P, Ma Z, Shi L, Xue Y, Guillen-Guillen I, Huang L, Hishida T, Liao H-K, Nuñez Delicado E, Rodriguez Esteban C, Guillen-Garcia P, Reddy P, Izipisua Belmonte JC (2021) **In vivo partial reprogramming of myofibers promotes muscle regeneration by remodeling the stem cell niche** *Nat Commun* **12**:3094
46. Wang L-L, Serrano C, Zhong X, Ma S, Zou Y, Zhang C-L (2021) **Revisiting astrocyte to neuron conversion with lineage tracing in vivo** *Cell* **184**:5465–5481
47. Wang S, Sengel C, Emerson MM, Cepko CL (2014) **A gene regulatory network controls the binary fate decision of rod and bipolar cells in the vertebrate retina** *Dev Cell* **30**:513–527
48. Wan J, Goldman D (2016) **Retina regeneration in zebrafish** *Curr Opin Genet Dev* **40**:41–47
49. Xiao D, Jin K, Qiu S, Lei Q, Huang W, Chen H, Su J, Xu Q, Xu Z, Gou B, Tie X, Liu F, Liu S, Liu Y, Xiang M (2021) **Regeneration of Ganglion Cells for Vision Restoration in Mammalian Retinas** *Front Cell Dev Biol* **9**:755544
50. Xie Y, Zhou J, Chen B (2022) **Critical examination of Ptbp1-mediated glia-to-neuron conversion in the mouse retina** *Cell Rep* **39**:110960
51. Yao K, Qiu S, Wang YV, Park SJH, Mohns EJ, Mehta B, Liu X, Chang B, Zenisek D, Crair MC, Demb JB, Chen B (2018) **Restoration of vision after de novo genesis of rod photoreceptors in mammalian retinas** *Nature* **560**:484–488
52. Yu J-H, Nam B-G, Kim M-G, Pyo S, Seo J-H, Cho S-R (2021) **Expression of Reprogramming Factor OCT4 Ameliorates Myelination Deficits and Induces Striatal Neuroprotection in Huntington's Disease** *Genes* **12** <https://doi.org/10.3390/genes12050712>

53. Zalc A, Sinha R, Gulati GS, Wesche DJ, Daszczuk P, Swigut T, Weissman IL, Wysocka J (2021) **Reactivation of the pluripotency program precedes formation of the cranial neural crest** *Science* **371** <https://doi.org/10.1126/science.abb4776>

Author information

Nguyet Le

Department of Neuroscience, Johns Hopkins University School of Medicine, Baltimore, United States

Sherine Awad

Department of Ophthalmology and Visual Sciences, University of Michigan School of Medicine, Ann Arbor, United States, Department of Cell and Developmental Biology, University of Michigan School of Medicine, Ann Arbor, United States, Michigan Neuroscience Institute, University of Michigan School of Medicine, Ann Arbor, United States

Isabella Palazzo

Department of Neuroscience, Johns Hopkins University School of Medicine, Baltimore, United States

Thanh Hoang

Department of Ophthalmology and Visual Sciences, University of Michigan School of Medicine, Ann Arbor, United States, Department of Cell and Developmental Biology, University of Michigan School of Medicine, Ann Arbor, United States, Michigan Neuroscience Institute, University of Michigan School of Medicine, Ann Arbor, United States

Seth Blackshaw

Department of Neuroscience, Johns Hopkins University School of Medicine, Baltimore, United States, Department of Ophthalmology, Johns Hopkins University School of Medicine, Baltimore, United States, Department of Neurology, Johns Hopkins University School of Medicine, Baltimore, United States, Institute for Cell Engineering, Johns Hopkins University School of Medicine, Baltimore, United States, Kavli Neuroscience Discovery Institute, Johns Hopkins University School of Medicine, Baltimore, United States
ORCID iD: [0000-0002-1338-8476](https://orcid.org/0000-0002-1338-8476)

For correspondence: sblack@jhmi.edu

Editors

Reviewing Editor

Xiaorong Liu

University of Virginia, Charlottesville, United States of America

Senior Editor

Lois Smith

Boston Children's Hospital, Boston, United States of America

Reviewer #1 (Public review):

Summary:

In this study, Le et al., aimed to explore whether AAV-mediated overexpression of Oct4 could induce neurogenic competence in adult murine Müller glia, a cell type that, unlike its counterparts in cold-blooded vertebrates, lacks regenerative potential in mammals. The primary goal was to determine whether Oct4 alone, or in combination with Notch signaling inhibition, could drive Müller glia to transdifferentiate into bipolar neurons, offering a potential strategy for retinal regeneration.

The authors demonstrated that Oct4 overexpression alone resulted in the conversion of 5.1% of Müller glia into Otx2+ bipolar-like neurons by five weeks post-injury, compared to 1.1% at two weeks. To further enhance the efficiency of this conversion, they investigated the synergistic effect of Notch signaling inhibition by genetically disrupting Rbpj, a key Notch effector. Under these conditions, the percentage of Müller glia-derived bipolar cells increased significantly to 24.3%, compared to 4.5% in Rbpj-deficient controls without Oct4 overexpression. Similarly, in Notch1/2 double-knockout Müller glia, Oct4 overexpression increased the proportion of GFP+ bipolar cells from 6.6% to 15.8%.

To elucidate the molecular mechanisms driving this reprogramming, the authors performed single-cell RNA sequencing (scRNA-seq) and ATAC-seq, revealing that Oct4 overexpression significantly altered gene regulatory networks. They identified Rfx4, Sox2, and Klf4 as potential mediators of Oct4-induced neurogenic competence, suggesting that Oct4 cooperates with endogenously expressed neurogenic factors to reshape Müller glia identity.

Overall, this study aimed to establish Oct4 overexpression as a novel and efficient strategy to reprogram mammalian Müller glia into retinal neurons, demonstrating both its independent and synergistic effects with Notch pathway inhibition. The findings have important implications for regenerative therapies as they suggest that manipulating pluripotency factors *in vivo* could unlock the neurogenic potential of Müller glia for treating retinal degenerative diseases.

Strengths:

- (1) Novelty: The study provides compelling evidence that Oct4 overexpression alone can induce Müller glia-to-bipolar neuron conversion, challenging the conventional view that mammalian Müller glia lacks neurogenic potential.
- (2) Technological Advances: The combination of Müller glia-specific labeling and modifying mouse line, AAV-GFAP promoter-mediated gene expression, single-cell RNA-seq, and ATAC-seq provides a comprehensive mechanistic dissection of glial reprogramming.
- (3) Synergistic Effects: The finding that Oct4 overexpression enhances neurogenesis in the absence of Notch signaling introduces a new avenue for retinal repair strategies.

Weaknesses:

- (1) In this study, the authors did not perform a comprehensive functional assessment of the bipolar cells derived from Müller glia to confirm their neuronal identity and functionality.
- (2) Demonstrating visual recovery in a bipolar cell-deficiency disease model would significantly enhance the translational impact of this work and further validate its therapeutic potential.

Comments on revisions:

The author answered all my questions and corrected the minor comments, so I have no more comments on the manuscript.

<https://doi.org/10.7554/eLife.106450.2.sa2>

Reviewer #2 (Public review):**Summary:**

The authors harness single cell RNAseq data from zebrafish and mice to identify Oct4 as a candidate driver of neurogenesis. They then use adeno-associated virus vectors to show that while Oct4 overexpression alone converts rare adult Müller glia (MG) to bipolar cells, it synergizes with Notch pathway inhibition to cause this neurogenesis (achieved by Cre-mediated knockout of Rbpj floxed allele). Importantly, they genetically lineage-mark adult MG using a GLAST-CreER transgene and a Sun-GFP reporter, so that any non-MG cells that convert can be identified unambiguously. This is crucial because several high-profile papers made erroneous claims using short promoters in the viral delivery vector itself to mark MG, but those promoters are leaky and mark other non-MG cell types, making it impossible to definitively state whether manipulations studied were actually causing neurogenesis, or were merely the result of expression in pre-existing neurons. Once the authors establish Oct4 + RbpjKO synergy they use snRNAseq/ATACseq to identify known and novel transcription factors that could play a role in driving neurogenesis.

Strengths:

The system to mark MG is stringent, so the authors are studying transdifferentiation, not artifactual effects due to leaky viral promoters. The synergy between Oct4 and Notch pathway blockade is notable. The single cell results add the potential involvement of new players such as Rfx4 in adult-MG-neurogenesis.

Weaknesses:

The revised version is clear and there are no major weaknesses.

Overall, the authors achieved what they set out to do, and have made new insights into how neurogenesis can be stimulated in MG. Ultimately, a major long-term goal in the field is to replace lost photoreceptors as this is most relevant to many human visual disorders, and while this paper (like all others before it) does not generate rods or cones, it opens new strategies to coax MG to form a related neuronal cell type. Their approach underscores the benefits of using a gold standard approach for lineage tracing.

<https://doi.org/10.7554/eLife.106450.2.sa1>

Author response:

The following is the authors' response to the original reviews

Public Reviews:**Reviewer #1 (Public review):****Summary:**

In this study, Le et al.. aimed to explore whether AAV-mediated overexpression of Oct4 could induce neurogenic competence in adult murine Müller glia, a cell type that, unlike its counterparts in cold-blooded vertebrates, lacks regenerative potential in mammals. The primary goal was to determine whether Oct4 alone, or in combination with Notch signaling inhibition, could drive Müller glia to transdifferentiate into bipolar neurons, offering a potential strategy for retinal regeneration.

The authors demonstrated that Oct4 overexpression alone resulted in the conversion of 5.1% of Müller glia into Otx2+ bipolar-like neurons by five weeks post-injury, compared to 1.1% at two weeks. To further enhance the efficiency of this conversion, they investigated the synergistic effect of Notch signaling inhibition by genetically disrupting Rbpj, a key Notch effector. Under these conditions, the percentage of Müller gliaderived bipolar cells increased significantly to 24.3%, compared to 4.5% in Rbpjdeficient controls without Oct4 overexpression. Similarly, in Notch1/2 double-knockout Müller glia, Oct4 overexpression increased the proportion of GFP+ bipolar cells from 6.6% to 15.8%.

To elucidate the molecular mechanisms driving this reprogramming, the authors performed single-cell RNA sequencing (scRNA-seq) and ATAC-seq, revealing that Oct4 overexpression significantly altered gene regulatory networks. They identified Rfx4, Sox2, and Klf4 as potential mediators of Oct4-induced neurogenic competence, suggesting that Oct4 cooperates with endogenously expressed neurogenic factors to reshape Müller glia identity.

Overall, this study aimed to establish Oct4 overexpression as a novel and efficient strategy to reprogram mammalian Müller glia into retinal neurons, demonstrating both its independent and synergistic effects with Notch pathway inhibition. The findings have important implications for regenerative therapies as they suggest that manipulating pluripotency factors in vivo could unlock the neurogenic potential of Müller glia for treating retinal degenerative diseases.

Strengths:

(1) Novelty: The study provides compelling evidence that Oct4 overexpression alone can induce Müller glia-to-bipolar neuron conversion, challenging the conventional view that mammalian Müller glia lacks neurogenic potential.

(2) Technological Advances: The combination of Muller glia-specific labeling and modifying mouse line, AAV-GFAP promoter-mediated gene expression, single-cell RNA-seq, and ATAC-seq provides a comprehensive mechanistic dissection of glial reprogramming.

(3) Synergistic Effects: The finding that Oct4 overexpression enhances neurogenesis in the absence of Notch signaling introduces a new avenue for retinal repair strategies.

Weaknesses:

(1) In this study, the authors did not perform a comprehensive functional assessment of the bipolar cells derived from Müller glia to confirm their neuronal identity and functionality.

(2) Demonstrating visual recovery in a bipolar cell-deficiency disease model would significantly enhance the translational impact of this work and further validate its therapeutic potential.

Response: We thank the Reviewer for their evaluation. We agree that functional analysis of Müller glia-derived bipolar cells is indeed important, but is beyond the current scope of the manuscript.

Reviewer #2 (Public review):

Summary:

The authors harness single-cell RNAseq data from zebrafish and mice to identify Oct4 as a candidate driver of neurogenesis. They then use adeno-associated virus vectors to show

that while Oct4 overexpression alone converts rare adult Müller glia (MG) to bipolar cells, it synergizes with Notch pathway inhibition to cause this neurogenesis (achieved by Cre-mediated knockout of Rbpj floxed allele). Importantly, they genetically lineage-mark adult MG using a GLAST-CreER transgene and a Sun-GFP reporter, so that any non-MG cells that convert can be identified unambiguously. This is crucial because several high-profile papers made erroneous claims using short promoters in the viral delivery vector itself to mark MG, but those promoters are leaky and mark other non-MG cell types, making it impossible to definitively state whether manipulations studied were actually causing neurogenesis, or were merely the result of expression in pre-existing neurons. Once the authors establish Oct4 + RbpjKO synergy they use snRNAseq/ATACseq to identify known and novel transcription factors that could play a role in driving neurogenesis.

Strengths:

The system to mark MG is stringent, so the authors are studying transdifferentiation, not artifactual effects due to leaky viral promoters. The synergy between Oct4 and Notch pathway blockade is notable. The single-cell results add the potential involvement of new players such as Rfx4 in adult-MG-neurogenesis.

Weaknesses:

The existing version is difficult to read due to an unusually high number of text errors (e.g. references to the wrong figure panels etc.). A fuller explanation for the fraction of non-MG cells seen in control scRNAseq assays is required, particularly because the neurogenic trajectory which is enhanced in the Oct4/Rbpj-KO context is also evident in the control retina. Claims regarding the involvement of transcription factors in adult neurogenesis (such as Rfx4) need to be toned down unless they are backed up with functional data. It is possible that such factors are important, but equally, they may have no role or a redundant role, and without functional tests, it's impossible to say one way or the other.

Overall, the authors achieved what they set out to do, and have made new insights into how neurogenesis can be stimulated in MG. Ultimately, a major long-term goal in the field is to replace lost photoreceptors as this is most relevant to many human visual disorders, and while this paper (like all others before it) does not generate rods or cones, it opens new strategies to coax MG to form a related neuronal cell type. Their approach underscores the benefits of using a gold-standard approach for lineage tracing.

We thank the Reviewer for their evaluation. We have made extensive changes to the manuscript to correct errors and modify discussion as recommended. These are detailed below in our point-by-point responses to specific recommendations to the authors.

Recommendations for the authors:

Reviewer #1 (Recommendations for the authors):

Minor corrections:

(1) In Figure 1C top GFAP-mCherry panel, two dim GFP + cells have colocalized with Otx2, is it caused by optic imaging thickness or some muller glia cells having the Otx2 expression?

This indeed reflects the effects of optic imaging thickness. Colocalization of Sun1-GFP and Otx2 is not observed when Z-stack images are examined in GlastCreER;Sun1-GFP retinas. This can also be appreciated by the fact that, in cases of apparent overlap of nuclear envelope-targeted Sun1 and Otx2, the sizes of the labeled areas differ. In cases of true expression

overlap, such as is seen following Oct4 overexpression, the labeled areas are the same size, or very nearly so.

Whether the Glast-CreERT2 x Rosa26-LSL-Sun1-GFP mouse line has cross-labeling with the Otx2+ bipolar cells, the author should image the mCherry ctrl sample with a thin optical imaging layer with a small pinhole for Z-stack to verify the co-labeling the GFP and Otx2 in mCherry ctrl sample.

Please see above. Since we first described this line (de Melo, et al. 2012), we have examined thousands of sections of GlastCreER;Sun1-GFP retinas, and have yet to see a single GFP-positive neuron. To avoid confusion, however, we have replaced these images with an additional image from a control mCherry-infected GlastCreER;Sun1-GFP retina processed for the same study.

In the middle upper panel, Oct4-mCherry group, the white arrows indicate the GFP colocalized with Otx2 signal, but seems not mCherry positive, by contrast, the neighbor cells have significant mCherry expression but no colocalization with Otx2. The GFAP promoter-Oct4-mCherry may have stopped expression after the Müller Glia cells were converted into Otx2+ bipolar cells, but is there any middle stage in which the Oct4mCherry and Otx2 co-expression? And after Müller glia to Bipolar conversion, why have Glast-CreERT2 driven GFP expressions not suppressed as GFAP promoter driven Oct4-mCherry? Could the author discuss this point?

We observed a significant number of Muller glia-derived cells expressing both Otx2 and weak mCherry signal. GFP expression is driven by the ubiquitous CAG promoter following Cre-dependent excision of a transcriptional stop cassette. We have modified the text to make this point explicit.

(2) In Figure S2b, the mouse is labeled with wild type; I assume it should be the same mouse line as Fig.1. Otherwise, the author should describe the source of the GFP signal.

“Wildtype” in this case refers to GlastCreER;Sun1-GFP controls, which as the Reviewer correctly points out, are not truly wildtype. The genotype of these animals is specified in all figure legends, and is now referred to as “control” rather than “wildtype” in the figures and main text throughout.

In Figure S2k and l, mCherry ctrl panel, the GFP+ cells looked co-labeling with Otx2, so again, is it the thicker optical imaging layer that caused overlapping vertically or the low specific of Müller Glia of the mouse line? Please describe the stars' meaning in Figure S2i,j in the figure legend. There are 2 figures labeled "n" of the quantification data.

This is, again, an example of the thicker optical imaging layer causing apparent overlap. We have previously demonstrated that the Sun1-GFP+ cells do not co-label with Otx2 in GFAP-mCherry AAV-injected control retinas (Le et al., 2022; Fig. 2C). The asterisks (*) indicate mouse-on-mouse vascular staining, which is now clarified in the figure legend. The 2 figures labeled ‘n’ have been relabeled as ‘m’ and ‘n’.

(3) In Figure 2c in the top panel, the Otx2 image was wrong; please replace it with the correct one.

We thank the Reviewer for spotting this error. This is an inadvertent duplication of the single-channel Otx2 staining for mCherry control sample. We have replaced this with the correct image.

(4) In Figure 3a, the *Rbpj*-cKO mouse line was used, but where was the GFP signal from? Please verify the mouse line you used in your work. The same question is also asked in Figure S3, S4b.

GlastCreER;*Rbpj*^{lox/lox};Sun1-GFP were used in Figure 3a. As now specified in the Methods and all figure legends, all mice used in this study carry both the GlastCreER and Sun1-GFP transgenes.

(5) In Figure S4c,d, and 5 wks time point, if the authors quantify the GFP+/Sox2- cells changing, it will be more helpful to understand the percentage of the Müller glia cells conversion to Bipolar cells compared to the Figure 2D, and can be as a supplement to the conclusion Müller to Bipolar conversion rather the Müller proliferation.

Sox2-/GFP+ cells are a measure of Müller glia to bipolar cell conversion that complements that of GFP+/Otx2+ cells. This is now clarified in the text. We also include quantification of Sox2-/GFP+ neurons at 5 weeks post-injury in Fig. S5b.

(6) In Figure S1b,c, there is a large portion of cells that are activated Müller glia after NMDA injury. Did the activated Müller glial cells lose their Müller glial identity? Between the loss of Müller glial identity and neuronal reprogramming, are there any markers that can be used to assess whether Müller glial cells are truly transdifferentiating into neurons rather than remaining in a reactive glial state or an intermediate phase?

Wildtype Müller glia progressively revert to resting state, and by 72 hours post-injury have already lost expression of *Klf4* and *Myc* (Hoang, et al. 2020), a point which is now specifically mentioned in the text. In GlastCreER;Sun1-GFP;*Nfia*/b/x^{lox/lox}; *Rbpj*^{lox/lox} Müller glia, reactive MG appear to largely convert to bipolar and amacrine-like cells, and it remains unclear if they eventually revert to a resting state (Le, et al. 2024).

Reviewer #2 (Recommendations for the authors):

This work demonstrates that Oct4 (Pou5f3) can induce neurogenesis in murine Müller glia (MG). Le et al start by showing that murine and zebrafish MG lack expression of Oct4 (Pou5f3) and its target Nanog. To assess the effect of Oct4 they first label adult MG with Sun1-GFP using tamoxifen-treated GlastCreER;Sun1-GFP mice, then later transduce in vivo with AAV vectors expressing mCherry alone or Oct4 + mCherry. Subsequently, they damage the retina with NMDA and assess the effects several weeks later. In Oct4+ cells at 2 weeks there is rare induction of the neural determinant Ascl1, down-regulation of the MG marker Sox2, induction of bipolar markers (Otx2, Scgn, Cabp5) but not amacrine (HuC/D) or rod (Nrl) markers. Combining Oct4 with

Notch inhibition (deleting floxed Rbpj) synergistically increases bipolar cell induction, with Otx2 staining rising to >20% of GFP-marked cells, and cells losing MG identity (loss of Sox2/9). EdU labeling was negligible suggesting direct trans-differentiation. Similar synergy was seen upon combining Oct4 expression with Notch1/2 double gene knockout. Attempts to combine Oct4 with Nfia, Nfib, and Nfix loss were unsuccessful as the GFAP promoter driving Oct4 in MG seems to require these three related transcription factors. scRNAseq confirmed the Oct4-overexpression/Rbpj-KO-driven increase in bipolar cells and decrease in MG cells and revealed that these manipulations may enhance bipolar cell genesis by repressing genes that define quiescent MG and enhancing expression of genes that define reactive MG and neurogenic cells. Finally, multiomic snRNA/scATAC-seq data was performed to assess the effect of Oct2 in wt or Rbpj null MG. This approach revealed that, as anticipated, more genes were up and down-regulated in the context of both manipulations vs Oct4 OE alone. Moreover, Oct4 and Rbpj KO reduced chromatin

accessibility at target motifs for transcription factors involved in MG identity/quiescence, while MGPCs showed elevated accessibility for neurogenic factors. The combination of Oct4 OE and Rbpj KO induces accessibility at various interesting TF sites that may contribute to the synergistic neurogenesis, including Rfx4, Klf4, Insm1, and others.

This is an interesting paper that adds to the growing literature on how neurogenesis can be induced in mammalian MG. The focus on Oct4 is interesting and the synergistic effects are striking and analyzed in some detail with scRNAseq and multiomic snRNA/scATACseq. The latter results provide useful new insight into transcriptional programs that may be critical in driving neurogenesis. Functional insight into these new candidates is not explored in this manuscript, but that's beyond the scope of the current work and forms the basis for new studies. There are some overreaching statements in the Discussion that need to be toned down, but apart from that and a long list of textual errors that need to be fixed, this paper is a valuable contribution to the field.

Major comments

There are numerous textual errors (some, but not all, examples are detailed in minor comments). It was difficult to follow this paper given the unusually high number of textual errors and the abbreviated legends. Greater attention should be paid to harmonizing the text with the figures and ensuring that the legends are correct and complete.

The manuscript has been proofread carefully and errors corrected.

The opening section of the scRNAseq data should outline briefly why sorting for GFP labeled cells purifies a significant fraction of non-MG cell types, despite the earlier claim, (which agrees with other publications), that GLAST-CreER transgene expression is highly specific to MG. Presumably, it mainly/totally reflects the co-purification of cells, cell fragments, and/or cell-free mRNA from other lineages. Is it also possible that a fraction (however small) of these cells reflect low-level spurious/temporary activation of GLAST-CreER expression in non-MG? The "contamination" is present despite the addition of the GFP sequence to the reference genome (as explained in Methods). They mention: "a clear differentiation trajectory connecting Muller glia, neurogenic Muller gliaderived progenitor cells (MGPCs), and differentiating amacrine and bipolar cells (Fig. 3b)". However, the same trajectory is evident in control mCherry samples, so one could argue that this trajectory is active in normal retina at some low rate, but that would/should equate to rare sun-GFP+ non-MG in controls. Are there any such cells, even extremely rarely, or is it truly 0%? At any rate, the authors need to raise these concerns and offer some explanation(s) at the start of their scRNAseq Results section. If there are really no such sun-GFP+ cells, the authors should comment on the presence of the apparent inactive trajectory in the Discussion.

Since we first described this line (de Melo, et al. 2012), we have examined thousands of sections of GlastCreER;Sun1-GFP retinas, and have yet to see a single GFP-positive neuron. We have also previously shown (Hoang, et al. 2020) that FACSbased isolation of GFP-positive cells from GlastCreER;Sun1-GFP yields a roughly thirty-fold enrichment of Muller glia, implying the presence of small numbers of contaminating neurons. We thereby conclude that the presence of small numbers of neurons (rods, cones, bipolar, and amacrine cells) in the control GlastCreER;Sun1-GFP represents contamination rather than low levels of glia-to-neuron conversion, particularly since we are unable to detect the expression of genes such as neurogenic bHLH factors or immature photoreceptor precursor-specific factors such as Prdm1 that indicate the presence of intermediate cell states. This is now addressed in the Results section related to both Figures 3 and 4.

In reference to other strategies to induce neurogenesis the authors make the claim that Oct4 is fundamentally different: "In these cases, Müller glia broadly upregulate proneural genes and/or downregulate Notch signaling. Oct4 instead induces expression of the neurogenic transcription factor Rfx4, which is not expressed in developing retina. It is likely that activation of this parallel pathway to neurogenic competence in part accounts for synergistic induction of neurogenesis seen in Rbpj-deficient Müller glia". First, all these strategies, including Oct4, seem to activate bHLH factors, so they have that in common and the authors should note that overlap. More seriously, without functional tests (e.g. KO Rfx4) the authors need to dial back the over-reaching statement that Rfx4 is the fundamental mechanism driving the Oct4 effect. They can certainly suggest that this is one possibility, but equally, Rfx4 may have very little or no effect on neurogenesis, or it could act redundantly with some of the other factors the authors uncovered. It's impossible to know without functional data, so they either need to add the functional data, or hold back on the strong one-sided and overreaching claim.

Since both Rfx4 expression and motif accessibility are selectively observed following Oct4 overexpression, and Rfx4 also has known neurogenic activity, we stand by our conclusion that it is a particularly strong candidate for mediating the neurogenic effects of Oct4 overexpression. However, the Reviewer is correct that in the absence of functional data, speculation about its function should be qualified. We have done this in the revised manuscript.

Minor comments

This sentence in the Results is confusing: "While expression of neurogenic bHLH factors driven by the Gfap promoter was rapidly silenced in Muller glia and activated in amacrine and retinal ganglion cells, Gfap-Oct4-mCherry remained selectively expressed in Muller glia but did not induce detectable levels of Muller glia-derived neurogenesis in the uninjured retina (Le et al., 2022)". The cited reference is at the end so it sounds like the Oct4 assay was performed in Le et al 2022, and there is no reference to a Figure for the Oct4 data in the current paper.

As stated here, in Le, et al. 2022, we did not observe any conversion of Sun1-GFP-positive Muller glia to neurons in the absence of injury. In the current study, we instead test whether NMDA-induced excitotoxicity induced glia to neuron conversion in Muller glia overexpressing Oct4. This is now made clear in the revised text.

There are many errors and omissions regarding Figure S2:

Figure S2a, b legend, and panels do not match. 2a should be a schematic of the strategy to label MG with Sun1-GFP using GLAST-Cre and a floxed Sun1-GFP allele, but that's missing and instead, the current 2a is a schematic of AAV vectors. It seems that the current 2b legend may describe the combination of the current 2a and 2b panels.

This has been corrected.

Figure S2: Asterisks label certain stained elements in the Oct4 labeled panels, but there is no explanation in the legend. Are these meant to indicate non-specific staining? If so, what is the evidence that the signal is non-specific?

These asterisks represent non-specific mouse-on-mouse vascular staining observed with the mouse monoclonal anti-Oct4 used in this study. This is now indicated in the figure legend.

The text refers to Ascl1 staining in Figure S2e,f, but it's S2g,h.

This has been corrected.

Re this: "While Sun1-GFP-positive cells infected with Oct4-mCherry mostly express the Muller glial marker Sox2 (Fig. S2a,b), from 2 weeks post-injury onwards a subset of GFP positive cells did not show detectable Sox2 expression (Fig. S2b, yellow arrows)". Figure S2a, b are schematic diagrams, not immunofluorescence. They probably mean Figure S2c, d.

This has been corrected.

Fig S2m is mislabeled "n".

This has been corrected.

There are probably other errors with this figure, but I mostly gave up at this point. The authors should go through the paper to find and correct any additional mistakes/omissions in the text and legends.

The manuscript has been carefully proofread and errors corrected.

The figure panels are not always mentioned in the order that they appear. There are many examples.

Figure panels are now mentioned in the order that they appear.

Several schematics use "d-18-14" to indicate "day -18 to -14". The former is at first uninterpretable or at best unclear (could mean day -18 to day 14), perhaps d -18 to -14, or d -18:-14 would be clearer.

This has been corrected.

Re: "AAV-infected wildtype Muller glia could be readily identified by selective expression of Oct4 (Fig. 4e). Wildtype Oct4-expressing Muller glia give rise to both small numbers of neurogenic MGPCs (Fig. 4b),". Figure 4E is labeled Pou5f1, but it would be helpful to avoid confusion by also indicating on the figure that Pou5f1 = Oct4; and Fig 4b does not indicate neurogenic MGPCs (perhaps they mean 4c).

This has been corrected.

Some parts of the Results are written in the present tense and should be in the past tense (for guidance: <https://www.nature.com/scitable/topicpage/effective-writing13815989/>).

Past tense is now used throughout.

Pit1 (Pou1f1) is referred to as a "close variant" of Oct4/Pou4f5, but this is unclear (e.g. variant could mean a splice variant from the same locus) and the term "paralogue" should be used.

"Paralogue" is now used in this context.

Re: "Infection with Oct4-mCherry vector induced both Oct4 (Fig. S5e) and Ascl1 (Fig. S5d) expression in Notch1/2-deficient Müller glia." Supplementary image 5d is the one depicting Oct4 and 5e is the one showing Ascl1. However, the reference is reversed.

This has been corrected.

<https://doi.org/10.7554/eLife.106450.2.sa0>